Journal of Machine Learning Research 1–63 Submitted 4/26

The Attribution Impossibility:
No Importance Ranking Is Faithful, Stable, and Complete Under Symmetry

Drake Caraker DRAKECARAKER@GMAIL.COM
Independent Researcher

Bryan Arnold PUREMATH86@GMAIL.COM
Independent Researcher

David Rhoads DRHOADS9@GMAIL.COM
Independent Researcher

Abstract

Importance rankings—of input features (SHAP) and internal components (activation patching)—change when you retrain the model. On Breast Cancer, 50 retrains produce 24 distinct top-3 rankings. Ten GPT-2-small models produce 32 distinct top-5 component rankings. When correlated elements have similar importance, the ranking is determined by the training seed, not by the data.

The root cause is a conflict among three desiderata: no importance ranking can be simultaneously faithful, stable, and complete when the Rashomon property holds. We characterize the **complete design space**: exactly two families of methods exist, and no third option is possible. DASH (the orbit average under the natural symmetry group) is the optimal equivariant estimator on the stable branch by classical invariant decision theory. For gradient-boosted stumps, the attribution ratio diverges as $1/(1-\rho^2)$; the qualitative pattern—GBDT diverges, Lasso infinite, random forests converge—holds across depths. At the component level, the G-invariant projection lifts agreement from $\rho = 0.277$ to 0.873 on GPT-2 (Cohen's $d = 19.2$) and from 0.55 to > 0.97 on TinyStories. Two different ablation methods converge after orbit averaging ($\rho > 0.97$ despite raw $\rho \approx 0.5$). The framework is verified in Lean 4 (368 theorems, 6 axioms, 0 `sorry`). The formalization caught two logical inconsistencies and one type mismatch that survived informal review.

Keywords: Attribution impossibility, faithful stable complete, Shapley value stability, mechanistic interpretability, Rashomon effect, ensemble explanation methods, formal verification (Lean 4)

Notation

Symbol	Meaning
$\varphi_j(f)$	Attribution (global importance) of feature j in model f
ρ	Pairwise correlation between collinear features within a group
T	Number of boosting rounds (trees) in a GBDT
M	Ensemble size (number of independently trained models)
$n_j(f)$	Split count (utilization count) of feature j in model f
$c(f)$	Proportionality constant: $\varphi_j(f) = c(f) \cdot n_j(f)$
P	Total number of features
L	Number of collinear groups
m	Number of features within a collinear group
j_1	First-mover: the feature selected at the root of the first tree
$\bar{\varphi}_j$	Dash consensus attribution: $\frac{1}{M}\sum_{i=1}^{M}\varphi_j(f_i)$
S	Ranking stability (expected Spearman correlation)
U	Within-group unfaithfulness
Δ_{jk}	Population attribution gap: $\lvert\mu_j - \mu_k\rvert$
σ_{jk}	Attribution noise: $\sqrt{\mathrm{Var}(\varphi_j(f) - \varphi_k(f))}$

1. Introduction

Train a gradient-boosted model on your data. Compute SHAP values. Retrain with a different random seed. The "most important feature" changes—in 68% of 77 public datasets. Train 10 GPT-2-small models from scratch. Run activation patching. The "most important attention head" changes—32 distinct top-5 rankings from 10 seeds. Before this work, practitioners assumed this was an engineering problem. We show it is a mathematical inevitability: faithfulness, stability, and completeness conflict under the Rashomon property. We characterize the *complete design space*: exactly two families of methods exist, Dash (the orbit average) is the optimal equivariant estimator on the stable branch, and no third option is possible. We provide quantitative bounds, a constructive resolution, and machine verification in Lean 4.

For symmetric elements (correlated features with similar importance, or heads within a layer), the flip rate approaches 1/2.

The classical multicollinearity concern is about estimation variance. Our result is qualitatively different: it concerns *ranking impossibility*. Even with perfect estimates, the ranking of symmetric features is determined by the training seed, not by the data.

A concrete example. Consider a credit model with annual income and debt-to-income ratio (typically $|\rho| \approx 0.7$–0.85 in lending data). Across 20 retrains with different seeds, SHAP can report income as the top driver roughly half the time and DTI the other half. A compliance report citing income as the primary adverse factor is unstable—the same model architecture, same data, different random seed. Under the Equal Credit Opportunity Act (ECOA), the lender's adverse action notice changes based solely on the training randomness.

Several recent results have established fundamental limits on feature attribution. Bilodeau et al. (2024) prove that completeness and linearity cannot coexist; Huang and Marques-Silva (2024) show SHAP can misrank features in Boolean domains; Srinivas and Fleuret (2019) show complete attributions cannot be weakly input-dependent; and Rao (2025) establishes

Kolmogorov complexity barriers to explainability. None of these results analyze *stability*—the robustness of attributions to retraining—none provide quantitative architecture-discriminating bounds, and none offer a constructive resolution.

This paper fills these gaps. Our result is not a criticism of SHAP, which correctly computes Shapley values for each model; rather, it reveals a fundamental limitation of the single-model paradigm under which any attribution method must operate. Our main result is the *Attribution Design Space Theorem* (§ 7, Figure 2): a complete characterization of what is achievable when explaining models with collinear features. The achievable set consists of exactly two families—faithful-complete methods (unstable, unfaithful to half the models) and ensemble methods like Dash (stable, with within-group ties)—and Dash is Pareto-optimal among unbiased aggregations. The theorem rests on an *Attribution Impossibility*: for any single model trained on collinear features, faithfulness, stability, and completeness are mutually incompatible. The impossibility has two layers. The first is model-agnostic: formalizing the Rashomon property (Rudin et al., 2024), we show symmetric features are necessarily ranked in opposite orders by different near-optimal models. The second is model-specific: we derive quantitative bounds on the attribution ratio (the factor by which the dominant feature is overweighted). For GBDT, this ratio follows $1/(1 - \alpha\rho^2)$ where α captures the per-tree signal fraction; for Lasso it is infinite at any $\rho > 0$; for neural networks it depends on the initialization; and for random forests it is $O(1/\sqrt{T})$—converging rather than diverging.

The impossibility parallels fairness: Chouldechova (2017) and Kleinberg et al. (2017) proved that calibration, balance, and equal false positive rates cannot coexist when base rates differ. Our result plays the same role for explainability—instability under collinearity is not an engineering deficiency but a mathematical inevitability—with direct implications for the EU AI Act (EU Parliament, 2024) (Art. 13(3)(b)(ii)), which requires disclosing "known and foreseeable circumstances" which may lead to risks to health, safety or fundamental rights.

The impossibility also admits a principled relaxation. Dash ensemble averaging breaks the sequential dependence that drives attribution concentration: for balanced ensembles, expected attributions are equitable across symmetric features with variance $O(1/M)$. Practitioners can achieve between-group faithfulness and stability by aggregating across the Rashomon set, sacrificing within-group completeness (reporting ties for symmetric features).

The proof is mechanically verified in Lean 4 using Mathlib (see § 14 for details). The core impossibility (Theorem 8) depends on zero axioms beyond the Rashomon property; the DASH equity result (Corollary 21) depends on the derived `attribution_sum_symmetric` theorem. The formalization also proved valuable beyond certification: during translation to Lean 4 we discovered two logical inconsistencies and one type mismatch in the original axiom system, echoing prior experiences where formalization uncovered subtle proof errors (Nipkow, 2009; Zhang et al., 2026).

Contributions.

1. **The Attribution Design Space Theorem** (§ 7): a complete characterization of achievable (stability, unfaithfulness, completeness) triples, showing the design space consists of exactly two families with Dash Pareto-optimal among unbiased aggregations on the ensemble branch. All other results are corollaries.

2. **The Attribution Impossibility** (Theorem 8): the base case of the Design Space Theorem—a model-agnostic impossibility requiring only the Rashomon property, with zero axiom dependencies in the Lean formalization.

3. **Architecture discrimination**: attribution ratios for GBDT ($1/(1-\rho^2)$ Lean-verified; $1/(1-\alpha\rho^2)$ empirically corrected), Lasso (∞), and random forests ($O(1/\sqrt{T})$, informal).

4. **Constructive relaxation via Dash**: ensemble averaging restores equitable attributions (derived for balanced ensembles) with variance $O(1/M)$, sacrificing within-group completeness.

5. **The symmetric Bayes dichotomy** (§ 8): a proof technique from invariant decision theory (Lehmann and Romano, 2005), demonstrated across three structurally distinct instances (feature attribution, model selection, and causal discovery under Markov equivalence) with different symmetry groups.

6. **Machine-verified proof**: formalized in Lean 4 (368 type-checked theorems from 6 axioms, 0 `sorry`; 139 with multi-step proofs of $\geq$5 tactic lines), publicly available at `https://github.com/DrakeCaraker/dash-impossibility-lean` (archived at `https://doi.org/10.5281/zenodo.19468379`). The axiom system is consistent: we construct an explicit model satisfying all 6 axioms simultaneously, both in Lean (`Consistency.lean`) and numerically.

The core impossibility proof is deliberately simple—a four-line contradiction from the Rashomon property. The mathematical contribution lies not in proof complexity but in identifying the right abstraction (the Rashomon property) and characterizing the complete achievable set (the Design Space Theorem). The quantitative depth comes from the architecture-specific bounds (§ 5) and the Pareto optimality proof (§ 6).

2. Related Work

Attribution impossibility results. Bilodeau et al. (2024) prove completeness and linearity cannot coexist; Huang and Marques-Silva (2024) show SHAP can misrank features in Boolean settings; Srinivas and Fleuret (2019) prove complete attributions cannot be weakly input-dependent; Rao (2025) establishes Kolmogorov complexity barriers. To our knowledge, our result is the first to simultaneously address cross-model *stability* as an impossibility, give quantitative architecture-discriminating bounds, and provide a constructive resolution with proved optimality. Decker et al. (2024) optimize aggregation across attribution *methods* for a single model; we aggregate across *models* via Dash. Jin et al. (2025) give constructive stability certificates; we give the impossibility that necessitates them. Noguer i Alonso (2025) derives information-theoretic limits on explanation complexity; our limits are about ranking consistency under feature correlation. The Bilodeau and Rashomon impossibilities are complementary: Bilodeau constrains methods satisfying linearity; ours constrains methods under collinearity (Table 1).

Table 1: Positioning relative to prior attribution impossibility results. These criteria reflect the specific contributions of the present work; each prior result makes distinct contributions not captured by this comparison.

Result	Stability?	Quantitative?	Resolution?	Formal?
Bilodeau et al. (2024)	No	No	No	No
Huang et al. (2024)	No	No	No	No
Srinivas et al. (2019)	No	No	No	No
Rao (2025)	No	No	No	No
Laberge et al. (2023)	Empirical	No	No	No
Herren & Hahn (2023)	Inference	No	No	No
Jin et al. (2025)	Certificates	No	No	No
This paper	**Impossibility**	**Yes**	**Dash**	**Lean 4**

Rashomon effect and model multiplicity. Laberge et al. (2023) find that attribution rankings across Rashomon sets are inherently partial orders; Rudin et al. (2024) advocate embracing model multiplicity. Our theorem shows the partial order is not a pragmatic choice but a mathematical necessity. Herren and Hahn (2023) develop statistical inference for feature rankings under model multiplicity; our contribution is to prove that the multiplicity is unavoidable under collinearity, not merely empirically common. Krishna et al. (2022) document the "disagreement problem" in explainable ML—different explanation methods produce different outputs for the same model—complementary to our result that the *same* method produces different outputs for different equivalent models.

Attribution sensitivity and fragility. Ghorbani and Zou (2019) demonstrate that neural network attributions are empirically fragile to small input perturbations. Hooker et al. (2021) show that unrestricted permutation-based importance forces extrapolation into out-of-distribution regions. Our instability arises from a different mechanism (model multiplicity, not input perturbation or extrapolation), but the practical consequence—unreliable explanations—is shared.

Foundational methods. Feature attribution via Shapley values (Shapley, 1953) was introduced to ML by Lundberg and Lee (2017). The model architectures we analyze were introduced by Breiman (2001) (random forests) and Friedman (2001) (gradient boosting); LIME (Ribeiro et al., 2016) provides an alternative local explanation framework.

Fairness impossibility. Chouldechova (2017) and Kleinberg et al. (2017) prove calibration, balance, and equal false positive rates cannot coexist. Our trilemma (faithfulness, stability, completeness) is structurally analogous.

Mechanistic interpretability and circuit multiplicity. **?** show that independently trained networks learn distinct internal representations for the same group operation. **?** find 85 distinct valid circuits with zero error for XOR. Our component-level results (Section 11) extend this line: the instability in circuit importance is mathematically inevitable under architectural symmetry, and the G-invariant projection provides a principled resolution.

Formal verification for ML. Nipkow (2009) formalized Arrow's theorem in Isabelle/HOL; Zhang et al. (2026) formalized statistical learning theory in Lean 4 (de Moura and Ullrich, 2021) using Mathlib (The mathlib Community, 2020). Our formalization is, to our knowledge, the first formally verified impossibility result in explainable AI.

3. Setup: Three Desiderata for Feature Rankings

We consider a supervised learning setting with P input features partitioned into L groups by their correlation structure. Within each group $\ell \in [L]$, features share a common pairwise correlation $\rho \in (0, 1)$; features in different groups are independent. Each group contains at least two members. This structure arises naturally in applied settings—e.g., multiple measurements of the same physical quantity, or one-hot encodings of related categories.

Models and attributions. A *model* f is trained from a random seed s via a deterministic training procedure: $f = \texttt{train}(s)$. For each feature $j \in [P]$, we observe a nonnegative attribution $\varphi_j(f) \geq 0$, representing the global importance of feature j in model f (e.g., mean absolute SHAP value, gain-based importance, or integrated gradients norm).

We require a proportionality axiom connecting attributions to model structure:

Axiom 1 (Proportionality) *For every model f, there exists a constant $c(f) > 0$ such that $\varphi_j(f) = c(f) \cdot n_j(f)$ for all $j \in [P]$, where $n_j(f)$ is the utilization count of feature j in model f (e.g., split count for tree ensembles, gradient norm for neural networks).*

This axiom holds exactly under the uniform-contribution model of Lundberg and Lee (2017), and approximately whenever per-split (or per-neuron) contributions are roughly homogeneous.

Sequential gradient boosting axioms. For gradient-boosted decision trees (GBDT) with T boosting rounds, we axiomatize the split-count structure induced by Gaussian conditioning under collinearity. Let j_1 denote the *first-mover*—the feature selected at the root of the first tree.

Axiom 2 (First-mover surjectivity) *For every group ℓ and every feature j in group ℓ, there exists a model f with $j_1(f) = j$.*

This follows from the data-generating process (DGP) symmetry: when features in a group have identical marginal distributions, sub-sampling and tie-breaking randomness ensure each can serve as first-mover.

Axiom 3 (Split counts) *For any model f with first-mover $j_1 \in$ group ℓ:*

$$n_{j_1}(f) = \frac{T}{2 - \rho^2}, \tag{1}$$

$$n_k(f) = \frac{(1 - \rho^2)\, T}{2 - \rho^2}, \quad k \in \textit{group}(\ell),\ k \neq j_1. \tag{2}$$

These are the leading-order split counts from the Gaussian conditioning argument: the first-mover absorbs the ρ-aligned signal component, leaving only the $(1 - \rho^2)$ residual for subsequent features. The formulas are verified algebraically by SymPy.

Table 2: Complete axiom inventory (6 axioms). Ten former axioms are now definitions or derived theorems.

Axiom	**Lean 4 name**	**Justification**
Model type	`Model`	Abstract type for trained models
First-mover function	`firstMover`	The dominant feature in each model
First-mover surjectivity	`firstMover_surjective`	DGP symmetry: each feature can be first-mover
Cross-group baseline	`crossGroupBaselineCore`	Split counts for features in other groups depend only on group indices
Proportionality constant	`proportionalityConstant`	$\varphi_j(f) = c \cdot n_j(f)$ with $c > 0$ uniform across models
Training measure	`modelMeasure`	Probability measure on Model (the training distribution)
Formerly axiomatized, now defined or derived (10 eliminated):		
Split count	`splitCount`	**Definition**: if-then-else on first-mover group membership
Split count (first-mover)	`splitCount_firstMover`	**Theorem**: derived from `splitCount` definition
Split count (non-first-mover)	`splitCount_nonFirstMover`	**Theorem**: derived from `splitCount` definition
Cross-group symmetry	`splitCount_crossGroup_symmetric`	**Theorem**: same `crossGroupBaselineCore` for same-group features
Cross-group stability	`splitCount_crossGroup_stable`	**Theorem**: `crossGroupBaselineCore` depends only on group indices
Attribution function	`attribution`	**Definition**: `proportionalityConstant` · `splitCount`
Global proportionality	`proportionality_global`	**Theorem**: derived from `attribution` definition
Boosting rounds	`numTrees`, `numTrees_pos`	**Fields** of `FeatureSpace` structure (T, $T > 0$)
Measurable space	`modelMeasurableSpace`	**Instance**: discrete σ-algebra ($\top$, works for any type)
Testing constant	`testing_constant`, `testing_constant_pos`	**Definition**: $C := 1/8$ from Tsybakov (2009); positivity by `norm_num`

Complete axiom inventory. Table 2 lists all property axioms in the formalization. The core impossibility theorem requires none of them—only the Rashomon property as hypothesis. The formalization uses 6 axioms total: 2 type declarations (`Model`, `firstMover`), 3 domain-specific property axioms (`firstMover_surjective`, `crossGroupBaselineCore`, `proportionalityConstant`), and 1 measure-theoretic axiom (`modelMeasure`). Ten former axioms—including the split-count structure, query-complexity constant, attribution function, and measurable space—are now definitions or derived theorems (see § 14).

Gaussian conditioning argument. The split count formulas (Axiom 3) are justified by the following argument. Consider two features X_j, X_k with correlation ρ and a GBDT with T boosting rounds. At each tree t, the root split selects the feature with the highest gain (variance reduction). Once feature j is selected as root, the available signal for k becomes:

$$X_k \mid X_j = X_k - \rho X_j, \qquad \mathrm{Var}(X_k \mid X_j) = 1 - \rho^2.$$

The conditional variance is reduced by factor $(1 - \rho^2)$ relative to the unconditional variance. In each subsequent boosting round, the first-mover's signal fraction is $1/(2 - \rho^2)$ (the probability that the first-mover's full signal exceeds the non-first-mover's residual signal in a two-feature competition). Summing over T trees: $n_{j_1} = T/(2-\rho^2)$ for the first-mover and $n_k = (1 - \rho^2)T/(2 - \rho^2)$ for non-first-movers.

The derivation assumes: (i) the root split captures the dominant signal direction (valid for stumps and low-depth trees); (ii) subsequent trees fit the residual from the first tree's split (the sequential boosting mechanism); (iii) the feature competition at each tree is approximately independent of previous trees' selections (valid when the learning rate η is small). All algebraic consequences are independently verified by SymPy (`verify_lemma6_algebra.py`).

Ranking desiderata. A *feature ranking* is a binary relation $\succ$ on $[P]$. We formalize three desiderata:

Definition 4 (Faithful) *A ranking $\succ$ is* faithful *to model f if $j \succ k$ whenever $\varphi_j(f) > \varphi_k(f)$.*

Definition 5 (Stable) *A ranking $\succ$ is* stable *if it does not depend on the choice of model f—that is, $\succ$ is a fixed relation applied identically to all models.*

Definition 6 (Complete) *A ranking $\succ$ is* complete *if for every pair $j \neq k$, either $j \succ k$ or $k \succ j$.*[1]

The first desideratum asks that the ranking reflect what the model actually learned; the second asks that it be reproducible across training runs; the third asks that it resolve all feature pairs.

Faithful, stable, complete: pick two (Figure 1).

Two explanation goals. Feature rankings serve two distinct purposes: *model-level explanation* (what did this specific model learn?) and *population-level explanation* (what does the model class learn?). Faithfulness is a model-level desideratum; stability is a population-level desideratum. The impossibility theorem (Theorem 8) characterizes the fundamental tension between these goals: a ranking cannot simultaneously be faithful to each individual model and stable across all models. Dash targets population-level explanation, sacrificing within-group completeness to achieve stability.

4. The Attribution Impossibility

4.1. The Rashomon Property

The central structural property driving the impossibility is that collinear features admit models ranking them in opposite orders.

1. Not to be confused with SHAP's "completeness" axiom ($\sum_j \varphi_j = f(x) - \mathbb{E}[f(X)]$). Our "complete" means *total*: the ranking decides every pair.

Definition 7 (Rashomon property) *A model class satisfies the* Rashomon property *if for every group ℓ and every pair of distinct features j, k in group ℓ, there exist models f, f' such that*

$$\varphi_j(f) > \varphi_k(f) \quad \textit{and} \quad \varphi_k(f') > \varphi_j(f').$$

The Rashomon property is a consequence of the Rashomon effect (Rudin et al., 2024; Fisher et al., 2019): collinear designs admit many near-optimal models (D'Amour et al., 2022), and within the set of good models, symmetric features are utilized in all possible orderings. Laberge et al. (2023) study consensus across such model sets and observe that feature rankings are inherently partial—a conclusion our theorem makes precise.

4.2. Main Result

Theorem 8 (Attribution Impossibility) *If a model class satisfies the Rashomon property (Definition 7), then no feature ranking can be simultaneously faithful, stable, and complete.*[2]

Proof Let j, k be distinct features in the same group ℓ. By the Rashomon property, there exist models f, f' with $\varphi_j(f) > \varphi_k(f)$ and $\varphi_k(f') > \varphi_j(f')$. Suppose for contradiction that $\succ$ is faithful, stable, and complete. By completeness, either $j \succ k$ or $k \succ j$. Without loss of generality, assume $j \succ k$. By stability, this relation holds for all models, including f'. But faithfulness applied to f' requires $k \succ j$ (since $\varphi_k(f') > \varphi_j(f')$), contradicting $j \succ k$. ■

When desirable properties conflict, relaxing completeness (accepting ties or partial orders) restores consistency. Dash achieves this by averaging attributions across the Rashomon set, producing $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features—a tie rather than an arbitrary ordering.

4.3. From Abstract to Concrete: Iterative Optimizers

The Rashomon property is not merely a theoretical possibility—it is a structural consequence of iterative optimization under collinearity.

Definition 9 (Iterative optimizer) *An* iterative optimizer *is a model class equipped with a dominant-feature function $d\colon \mathcal{F} \to [P]$ satisfying:*

1. ***Dominance:*** *For every model f, group ℓ, and feature $k \in group(\ell)$ with $k \neq d(f)$, if $d(f) \in group(\ell)$ then $\varphi_k(f) < \varphi_{d(f)}(f)$.*
2. ***Surjectivity:*** *For every group ℓ and feature $j \in group(\ell)$, there exists f with $d(f) = j$.*

Proposition 10 *Every iterative optimizer satisfies the Rashomon property, and therefore the Attribution Impossibility (Theorem 8) holds.*

Proof Let j, k be distinct features in group ℓ. By surjectivity, there exist models f, f' with $d(f) = j$ and $d(f') = k$. By dominance, $\varphi_j(f) > \varphi_k(f)$ and $\varphi_k(f') > \varphi_j(f')$. This is exactly the Rashomon property. ■

2. Both versions are Lean-verified: `attribution_impossibility` uses a biconditional ($j \succ k$ iff $\varphi_j(f) > \varphi_k(f)$); `attribution_impossibility_weak` uses the weaker implication (Definition 4) plus antisymmetry.

4.4. Rashomon Inevitability Under Symmetry

The Rashomon property is not merely a theoretical possibility for symmetric model classes—it is inevitable.

Definition 11 (Permutation closure) *A model class $\mathcal{F}$ is* permutation-closed *within group ℓ if for any $f \in \mathcal{F}$ and any permutation π of features within group ℓ, the composed model $f \circ \pi \in \mathcal{F}$.*

Theorem 12 (Rashomon from symmetry) *Let the DGP be symmetric within group ℓ (permuting features $j \leftrightarrow k$ leaves the population loss invariant). Let $\mathcal{F}$ be permutation-closed within group ℓ. Then for any model $f \in \mathcal{F}$ with $\varphi_j(f) \neq \varphi_k(f)$, the permuted model $f' = f \circ \pi_{jk}$ satisfies: (1) $L(f') = L(f)$ (same population loss), and (2) $\varphi_k(f') = \varphi_j(f)$ and $\varphi_j(f') = \varphi_k(f)$ (attributions swap). In particular, if $\varphi_j(f) > \varphi_k(f)$, then $\varphi_k(f') > \varphi_j(f')$, and the Rashomon property holds.*

Proof By DGP symmetry, the joint distribution of (X, Y) is invariant under permutation of features j and k within group ℓ. Therefore $L(f') = L(f \circ \pi_{jk}) = L(f)$, since the population loss depends only on the joint distribution. By construction, f' applies f to permuted inputs, so its reliance on feature k equals f's reliance on feature j: $\varphi_k(f') = \varphi_j(f)$ and vice versa. ■

The permutation closure condition is satisfied by any model class without built-in feature ordering: neural networks (permute input neurons), gradient-boosted trees (permute split features), Lasso (permute covariates), and random forests. It fails only for model classes with hard-coded feature preferences.

Definition 13 (Non-degeneracy) *Features j, k are* non-degenerate *if there exists at least one model f with $\varphi_j(f) \neq \varphi_k(f)$.*

Non-degeneracy is an existential condition requiring only one model that distinguishes the features—trivially satisfied in practice. Unlike a probability-1 claim, it needs no measure-theoretic machinery.

Theorem 14 (Rashomon inevitability) *Let $\mathcal{F}$ be permutation-closed under a symmetric DGP. If features j, k are non-degenerate, then the Rashomon property holds: there exist $f, f' \in \mathcal{F}$ with $\varphi_j(f) > \varphi_k(f)$ and $\varphi_k(f') > \varphi_j(f')$.*

Proof By non-degeneracy, there exists f with $\varphi_j(f) \neq \varphi_k(f)$; WLOG $\varphi_j(f) > \varphi_k(f)$. By Theorem 12, take $f' = f \circ \pi_{jk}$: this has the same loss and $\varphi_k(f') > \varphi_j(f')$. Both directions are realized. Machine-verified: `rashomon_inevitability` in `RashomonInevitability.lean` (zero axiom dependencies). ■

Remark 15 *The only escapes are: (i) perfect degeneracy ($\varphi_j = \varphi_k$ for ALL models), (ii) a model class that is not permutation-closed, or (iii) a DGP that is not symmetric within the group.*

5. Quantitative Bounds by Model Class

The Attribution Impossibility (Theorem 8) is qualitative. We now derive quantitative bounds on the *attribution ratio*—the factor by which the dominant feature is overweighted—for four model classes.

5.1. Gradient Boosting: Divergent Violation

Lemma 16 (Split gap) *For a GBDT model f with first-mover j_1 in group ℓ, and any $k \neq j_1$ in the same group,*

$$n_{j_1}(f) - n_k(f) = \frac{\rho^2 T}{2-\rho^2} \;\geq\; \tfrac{1}{2}\rho^2 T.$$

Proof By Axiom 3, $n_{j_1} - n_k = T/(2-\rho^2) - (1-\rho^2)T/(2-\rho^2) = \rho^2 T/(2-\rho^2)$. Since $2-\rho^2 \leq 2$ for $\rho \in (0,1)$, we have $\rho^2 T/(2-\rho^2) \geq \rho^2 T/2$. ■

Theorem 17 (Attribution ratio — gradient boosting) *For any GBDT model f with first-mover j_1 in group ℓ and any non-first-mover k in the same group,*

$$\frac{\varphi_{j_1}(f)}{\varphi_k(f)} = \frac{1}{1-\rho^2}.$$

Under full signal capture (α=1), this ratio diverges: $1/(1-\rho^2) \to \infty$ as $\rho \to 1^-$.

Proof By Axiom 1, $\varphi_{j_1}/\varphi_k = n_{j_1}/n_k$. Substituting Axiom 3:

$$\frac{n_{j_1}}{n_k} = \frac{T/(2-\rho^2)}{(1-\rho^2)T/(2-\rho^2)} = \frac{1}{1-\rho^2}.$$

Writing $1-\rho^2 = (1-\rho)(1+\rho)$, we see that $1/(1-\rho^2) \to +\infty$ as $\rho \to 1^-$. ■

For finite-depth trees, the effective signal capture $\alpha < 1$ yields a corrected ratio $1/(1-\alpha\rho^2)$; for stumps $\alpha \approx 2/\pi$ (R^2=0.89; Figure 5a).

The $\alpha = 2/\pi$ derivation. The value $\alpha \approx 2/\pi$ has a clean theoretical derivation from quantization theory.

Proposition 18 *For $X \sim \mathcal{N}(0,\sigma^2)$, the optimal binary quantizer (split at $x=0$) captures fraction $2/\pi$ of the variance:* $\mathrm{Var}(\hat{X}) = (2/\pi)\sigma^2$.

Proof The optimal two-level quantizer of a symmetric distribution splits at the median ($x=0$ for Gaussians). The quantized signal is $\hat{X} = \mathbb{E}[X \mid X>0] \cdot \mathbf{1}_{X>0} + \mathbb{E}[X \mid X \leq 0] \cdot \mathbf{1}_{X\leq 0}$. By symmetry, $\mathbb{E}[X \mid X>0] = \sigma\sqrt{2/\pi}$ (the mean of a half-normal distribution). Thus $\hat{X}$ takes values $\pm\sigma\sqrt{2/\pi}$ each with probability 1/2, giving $\mathrm{Var}(\hat{X}) = (2/\pi)\sigma^2$. ■

In each boosting round, a stump makes one binary split on the selected feature, capturing $2/\pi$ of that feature's remaining signal variance. The fitted value ($\alpha \approx 0.60$) is below $2/\pi = 0.637$; two error sources explain the gap: (1) empirical vs. population split location (negligible, $\Delta\alpha_1 \approx 0.0009$ for $n = 2000$), and (2) non-Gaussian residuals after the first boosting round (dominant, accounting for ≈ 0.036).

Table 3: Within-group split count ratio by depth and ρ (η=1.0).

	ρ=0.3	ρ=0.5	ρ=0.7	ρ=0.9	ρ=0.95	Fitted α
Depth 1 (stumps)	1.28	1.32	1.42	1.99	2.16	0.60 ($\approx 2/\pi$)
Depth 3	1.19	1.26	1.24	1.30	1.32	0.30
Depth 6	1.65	1.67	1.74	1.90	2.03	0.60
Depth 10	2.23	2.25	2.28	2.49	2.62	—
Theory (α=1)	1.10	1.33	1.96	5.26	10.26	1.00

Depth dependence. Table 3 reports the within-group split count ratio across tree depths and correlation levels.

At depth 3, each tree has up to 7 leaves and uses multiple features per tree. Internal splits distribute split counts across group members, counteracting the root split's first-mover advantage. The ratio is nearly ρ-independent (≈ 1.3), explaining the low fitted $\alpha = 0.30$. At depth 10, the root split cascades: once the first-mover is selected at the root, subsequent splits on the same feature are more likely at deeper levels (residuals retain first-mover signal). The baseline ratio (≈ 2.2 even at $\rho = 0.3$) is high because deep trees overfit to the first-mover's signal. The $1/(1-\alpha\rho^2)$ model does not fit well for depth 10 (the ratio has a large ρ-independent component).

Practitioner guidance. Depth 3 gives the lowest attribution instability among standard configurations. For users prioritizing explanation stability, shallow ensembles (depth 3) with Dash consensus ($M \geq 5$) provide the best stability–accuracy tradeoff.

Proportionality validation. The proportionality axiom holds with CV≈0.35 for stumps (the idealized theoretical setting) and CV≈0.66 for depth-6 trees on Breast Cancer. The quantitative ratio $1/(1-\rho^2)$ is an order-of-magnitude prediction, refined by the α-correction to $1/(1-\alpha\rho^2)$ with $R^2 = 0.89$. The core impossibility (Theorem 8) does not use this axiom. All quantitative bounds ($1/(1-\rho^2)$, ensemble size $M_{\min}$, attribution ratio) should be interpreted as order-of-magnitude predictions, not exact formulas.

Error analysis for $\alpha = 2/\pi$. The fitted $\alpha = 0.60$ (empirical) vs. $2/\pi = 0.637$ (theory) represents a gap of $\Delta\alpha = 0.037$. Two error sources: (1) empirical vs. population split location (negligible, $\Delta\alpha_1 \approx 0.0009$ for $n = 2000$): the optimal split of $X \sim \mathcal{N}(0, \sigma^2)$ at δ instead of 0 gives $\alpha_1(n) = (2/\pi)(1 - \pi(\pi-2)/(2n) + O(n^{-2}))$; (2) non-Gaussian residuals after the first boosting round (dominant, accounting for ≈ 0.036): residuals are a location-shifted mixture with excess kurtosis $\kappa = O(\eta^2 c^2/\sigma^2)$, which accumulates across boosting rounds. The impossibility theorem does not depend on α at all; the quantitative ratio uses α to predict the *severity* of the violation.

Stability bound. When two models f, f' have different first-movers within the same group of size m, the within-group rank reshuffling bounds the Spearman correlation:

$$\rho_S(f, f') \;\leq\; 1 - \frac{3m^2}{P^3 - P}.$$

Exact flip rate for GBDT.

Proposition 19 (Exact GBDT pairwise flip rate) *For features j, k in the same group of size m under the first-mover model with DGP symmetry (each feature equally likely to be first-mover):*

(i) The probability that a random model ranks $j > k$ is exactly $1/m$.

(ii) The tie probability (both non-first-movers with identical split counts) is $(m-2)/m$.

(iii) The flip rate across two independent models (excluding ties) is $2/m^2$.

(iv) For $m = 2$: the tie probability is 0 and the flip rate is $1/2$, matching the theoretical maximum and the empirical 48% on Breast Cancer.

(v) If ties are broken uniformly at random, the effective flip rate is exactly $1/2$ for all $m \geq 2$.

Proof By first-mover surjectivity and DGP symmetry, each feature in a group of size m is first-mover with probability $1/m$.

Part (i). Feature j is ranked above k when j is first-mover (probability $1/m$): the first-mover has split count $T/(2-\rho^2)$ vs. $(1-\rho^2)T/(2-\rho^2)$ for non-first-movers, so dominance holds.

Part (ii). When the first-mover is some feature $\ell \neq j, k$ (probability $(m-2)/m$), both j and k are non-first-movers with identical split count $(1-\rho^2)T/(2-\rho^2)$. By proportionality, $\varphi_j(f) = \varphi_k(f)$: a perfect tie.

Part (iii). A flip requires one model to rank $j > k$ and the other $k > j$: flip $= 2 \cdot (1/m) \cdot (1/m) = 2/m^2$.

Part (iv). At $m = 2$: flip $= 2/4 = 1/2$; tie probability $= 0$.

Part (v). When ties are broken uniformly, $\Pr[j > k \mid \text{tie}] = 1/2$. The effective $\Pr[j > k] = 1/m + (1/2)(m-2)/m = 1/2$ by algebra. Two independent draws disagree with probability $2 \cdot (1/2) \cdot (1/2) = 1/2$. ■

For $m = 2$ with exact DGP symmetry ($\mu_j = \mu_k$), the flip rate is $1/2$. For $m > 2$ without tie-breaking, most model pairs produce ties, with flips at rate $2/m^2$. But once ties are broken (as required for a complete ranking), symmetry forces the flip rate back to $1/2$. When DGP symmetry holds exactly, the impossibility is equally severe for every within-group pair, regardless of m.

5.2. Lasso: Infinite Violation

The ℓ_1 penalty selects exactly one feature from each correlated group and zeros all others. Lasso is an iterative optimizer with $d(f) =$ the selected feature; surjectivity holds because regularization-path tie-breaking can make any feature the selected one. The attribution ratio is therefore infinite at *any* $\rho > 0$.

5.3. Neural Networks: Conditional Violation

Neural networks exhibit symmetry breaking analogous to the GBDT first-mover effect. Surjectivity follows from the symmetry of standard initializations (Kaiming, Xavier). The attribution ratio is architecture-dependent, but the impossibility holds conditionally whenever training dynamics produce a dominant feature per group.

Table 4: Attribution ratio bounds by model class. T denotes the number of boosting rounds (or trees), ρ the within-group correlation.

Model class	Ratio	Scaling with T	Mechanism
Gradient boosting	$1/(1-\alpha\rho^2)$	Increases with ρ	Sequential residuals
Lasso	∞	Constant	Hard selection
Neural network	Model-dependent	Architecture-dependent	Init. symmetry breaking
Random forest[3]	$1+O(1/\sqrt{T})$	$\to 1$ as $T\to\infty$	Independent trees

5.4. Random Forests: Bounded Violation (Contrast)

Because each tree trains independently on a bootstrap sample, there is no shared residual stream. The expected split count is T/m for every feature in a group, with $O(\sqrt{T})$ deviations, giving

$$\frac{\varphi_{j_1}(f)}{\varphi_k(f)} = 1 + O(1/\sqrt{T}) \;\to\; 1 \quad \text{as } T \to \infty.$$

Sequential residual-sharing creates divergent inequity; independent aggregation creates convergent equity—the structural distinction Dash exploits.

Summary. Table 4 collects the quantitative bounds.

6. Resolution: Ensemble Attribution via Dash

The impossibility arises from sequential dependence: a single model's iterative optimization creates a dominant feature, and different random seeds create different dominant features. Dash addresses this by averaging attributions across an ensemble of M independently trained models, relaxing completeness in exchange for stability.

Definition 20 (Dash consensus attribution) *Given models $f_1, \ldots, f_M$ trained from independent seeds, the* consensus attribution *for feature j is $\bar{\varphi}_j = \frac{1}{M}\sum_{i=1}^{M}\varphi_j(f_i)$. An ensemble is* balanced *if, within each group, every feature serves as first-mover in exactly M/m models.*

Dash consensus vs. pipeline. Dash consensus (Definition 20) is the core mathematical operation, proved optimal in Theorems 22 and 25. The theoretical guarantees apply to any i.i.d. ensemble averaging, including simple seed averaging. A companion paper implements Dash as a full pipeline with diversity enforcement (hyperparameter variation, MaxMin model selection) and exploratory diagnostics (FSI, IS Plot) that improve finite-sample equity beyond seed averaging alone. For large M, seed averaging achieves approximate balance via the law of large numbers; diversity enforcement accelerates this convergence.

Corollary 21 (Dash achieves equity (axiom-derived)) *For a balanced ensemble and any two features j, k in the same group ℓ, $\bar{\varphi}_j = \bar{\varphi}_k$.*

Proof By DGP symmetry, permuting features $j \leftrightarrow k$ leaves the joint distribution invariant. For a balanced ensemble, each feature serves as first-mover equally often, so the summed split counts satisfy $\sum_i n_j(f_i) = \sum_i n_k(f_i)$. When the proportionality constant c is uniform across models, this gives $\sum_i \varphi_j(f_i) = \sum_i \varphi_k(f_i)$. ∎

Progressive Dash. For cost-sensitive deployments, we propose progressive Dash: start with M=5 models; for pairs with $Z > 3.0$, declare stable; for $Z < 1.96$, flag as unstable; for borderline pairs, train 5 more and repeat. Average-case cost: $\sim 8\times$ (not $25\times$). (The Z-thresholds 1.96 and 3.0 are fixed-sample values; a fully sequential design should use Pocock- or O'Brien-Fleming-corrected boundaries to control familywise error. We present this as a practical heuristic, not a formally analyzed procedure.)

Dash does not change predictions. Dash changes explanations, not predictions. The model's output is unchanged; only the accompanying explanation differs.

Between-group stability. Under independence of seeds, $\mathrm{Var}(\bar{\varphi}_j) = \mathrm{Var}(\varphi_j)/M$, which vanishes as $M \to \infty$.

Within-group completeness. Since $\bar{\varphi}_j = \bar{\varphi}_k$ for same-group features, the consensus ranking produces a *tie* rather than an arbitrary ordering. The tie is not a deficiency—it faithfully reflects the DGP symmetry.

6.1. DASH Pareto Optimality

Theorem 22 (DASH Pareto Optimality) *Let $f_1, \ldots, f_M$ be i.i.d. models with $\mu_j := \mathbb{E}[\varphi_j(f)]$ and $\sigma_j^2 := \mathrm{Var}(\varphi_j(f)) < \infty$.*

Part I (Lower bounds). *For any attribution aggregation method A:*

(a) If A is faithful to each individual model, then for symmetric features j, k, $U_{jk}(A) = 1/2$.

(b) For any unbiased linear *estimator $\hat{\mu}_j$ based on $f_1, \ldots, f_M$, $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ (Cauchy–Schwarz lower bound via Titu's lemma; Lean-verified via `sum_squares_ge_inv_M`); the extension to all unbiased estimators follows from the Rao–Blackwell theorem (standard; not formalized in Lean). In the parametric case, this coincides with the Cramér–Rao bound.*

Part II (Dash achieves the bounds).

(c) $U_{jk}(\mathrm{Dash}) = 0$ for symmetric features in balanced ensembles.

(d) $\mathrm{Var}(\bar{\varphi}_j) = \sigma_j^2/M$ (matching the Cauchy–Schwarz bound).

(e) For between-group features with gap Δ_{jk}: $S_{jk}(\mathrm{Dash}) \geq 1 - \exp(-M\Delta_{jk}^2/(2\sigma_{jk}^2))$.

Part III (Pareto optimality). *Among all methods achieving $U_{jk} = 0$ (ties for symmetric features), no method achieves strictly higher between-group stability than Dash for the same ensemble size M.*

Proof Part I(a): By DGP symmetry, $\Pr[\varphi_j(f) > \varphi_k(f)] = \Pr[\varphi_k(f) > \varphi_j(f)] = 1/2$. Any fixed ranking disagrees with exactly half the models (Theorem 26).

Part I(b): By the Cauchy–Schwarz inequality (Titu's lemma), for any weights w_i with $\sum w_i = 1$: $\sum w_i^2 \sigma_j^2 \geq \sigma_j^2/M$, with equality iff $w_i = 1/M$ for all i. Thus $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ for any unbiased linear estimator; Lean-verified via `sum_squares_ge_inv_M`. In the parametric case, the same bound follows from the Cramér–Rao bound with Fisher information M/σ_j^2.

Part II(c): By Corollary 21, $\bar{\varphi}_j = \bar{\varphi}_k$ for symmetric features in balanced ensembles. Dash reports a tie; since no ordering is asserted, $U = 0$.

Part II(d): For i.i.d. random variables, $\mathrm{Var}(\frac{1}{M}\sum_{i=1}^{M} X_i) = \mathrm{Var}(X_1)/M$. This matches Part I(b), so Dash achieves the Cauchy–Schwarz variance bound (Lean-verified via `sum_squares_ge_inv_M`; in the parametric case, this coincides with the Cramér–Rao bound). Moreover, since the sufficient statistic for μ_j from i.i.d. observations is $\sum \varphi_j(f_i)$, and $\bar{\varphi}_j$ is a function of this statistic, the Rao–Blackwell theorem (standard; not formalized in Lean) guarantees Dash dominates any estimator that is not a function of the sufficient statistic.

Part II(e): By Hoeffding's inequality (or the Gaussian approximation for large M): $\Pr[\bar{\varphi}_j < \bar{\varphi}_k] \leq \exp(-M\Delta_{jk}^2/(2\sigma_{jk}^2))$. For the Gaussian approximation (valid by CLT for $M \geq 30$): $\Pr[\bar{\varphi}_j < \bar{\varphi}_k] = \Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk})$.

Part III: Unfaithfulness $U = 0$ is already minimal. Stability S_{jk} is a monotone decreasing function of $\mathrm{Var}(\bar{\varphi}_j - \bar{\varphi}_k)$. By Part I(b), no unbiased method achieves lower variance than σ_j^2/M. Dash achieves this bound. For biased methods: any method with bias b_j has $\mathrm{MSE} = \mathrm{Var}(\hat{\varphi}_j) + b_j^2$; a bias of magnitude $|b_j - b_k| > \Delta_{jk}$ would invert the true ordering entirely. ■

Comparison with nonlinear aggregations. The sample median has asymptotic relative efficiency $2/\pi \approx 0.637$ relative to the mean (requiring $\pi/2 \approx 1.57$ times as many models). The 5%-trimmed mean achieves ARE $= 0.992$ with modest robustness. For standard ML training procedures, Dash (the mean) is strictly more efficient.

Connection to invariant decision theory. The DASH optimality has a deeper connection to classical invariant decision theory. The data-generating process is invariant under the permutation group S_m acting on within-group features. By the Hunt–Stein theorem (Lehmann & Romano, 2005, Theorem 9.3.3; for finite groups, the result follows from the complete class theorem), for a finite group, the best equivariant estimator is the Pitman estimator—the average over the group orbit (Lehmann & Romano, 2005, §6.7). For the permutation group acting on m features, this is exactly the sample mean of attributions, which is DASH. Hunt–Stein gives optimality among *all* estimators (not just unbiased ones), and directly explains why DASH produces ties: the group average is constant on orbits.

Best-approximation theorem. A stronger geometric optimality: the orbit average Rv is the *closest G-invariant vector* to v in L^2 norm. For any G-invariant w: $\|v - Rv\| \leq \|v - w\|$. The proof is the Pythagorean decomposition: $v = Rv + (v - Rv)$ where $Rv \in V^G$ and $(v - Rv) \perp V^G$. Then $\|v-w\|^2 = \|v - Rv\|^2 + \|Rv - w\|^2 \geq \|v - Rv\|^2$. This establishes that Dash minimizes information loss among *all* stable methods—not just unbiased or linear ones. Machine-verified: `reynolds_best_approximation`.

6.2. DASH Robustness and Breakdown Point

Dash (the sample mean) is the minimum-variance unbiased estimator, but it has breakdown point zero: a single adversarial or corrupted model can shift the consensus attribution arbitrarily. We compare robustness properties:

Table 5: Flip rate comparison: Dash vs. alternative stabilization methods. Synthetic Gaussian data (P=10, m=5, N=1000, 20 within-group feature pairs). 95% bootstrap CIs in brackets.

Method	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$	Pairs resolved
Single model	16.0% [13.8, 18.1]	16.4% [14.3, 18.5]	18.7% [16.4, 21.1]	40/40
Bootstrap SHAP	16.2% [13.8, 18.2]	16.1% [14.2, 18.0]	18.8% [16.4, 21.4]	40/40
Subsampled SHAP	16.3% [13.8, 18.5]	16.1% [14.2, 17.9]	18.8% [16.5, 21.3]	40/40
Dash (M=25)	**3.8%** [2.8, 4.9]	**3.1%** [2.4, 3.9]	**3.7%** [2.7, 4.7]	40/40
CI SHAP	0.2% [0.1, 0.4]	0.8% [0.6, 1.1]	3.1% [2.1, 4.6]	∼22/40

Method	ARE	Breakdown point	Recommendation
Mean (Dash)	1.000	0%	Standard use
5%-trimmed mean	0.992	5%	Moderate robustness
Median	0.637	50%	Adversarial settings

For production settings with potential model contamination (corrupted training data, adversarial seeds, or hardware failures producing degenerate models), the 5%-trimmed mean provides near-optimal efficiency (ARE = 0.992, requiring only 0.8% more models) with meaningful robustness. The median requires $\pi/2 \approx 57\%$ more models but tolerates up to 50% corrupted models. For $M = 25$: the 5%-trimmed mean discards the single most extreme model in each tail ($M_{\text{eff}} = 23$), a negligible cost. For adversarial robustness, the median with $M = 40$ achieves comparable stability to Dash with $M = 25$ while tolerating up to 20 corrupted models.

6.3. Empirical Comparison with Alternatives

We compare Dash against four alternative stabilization approaches across three correlation levels ($\rho \in \{0.5, 0.7, 0.9\}$), with 10 independent trials and bootstrap 95% confidence intervals on within-group flip rates. (With $n = 10$ trials, bootstrap confidence intervals should be interpreted cautiously; $n \geq 30$ would provide more reliable coverage.)

Bootstrap and subsampled SHAP address SHAP *estimation* noise (a single-model phenomenon) but not the Rashomon effect (a multi-model phenomenon). They were not designed for multi-model instability, so their flip rates matching the single-model baseline is expected rather than informative. They are included to confirm that estimation-noise methods do not incidentally address model-multiplicity noise. Only Dash trains multiple models, achieving a 4–5× reduction.

CI SHAP achieves lower flip rates than Dash by refusing to rank pairs whose confidence intervals overlap—CI SHAP resolves 50% of within-group pairs at $\rho = 0.5$, 56% at $\rho = 0.7$, and 63% at $\rho = 0.9$ (average ≈56%). This comparison is not apples-to-apples: CI SHAP's low flip rate applies only to resolved pairs, while Dash's 3–4% applies to all 40 pairs. CI SHAP exemplifies the impossibility theorem in action: it trades completeness for stability. At $\rho = 0.9$, its flip rate converges toward Dash's (3.1% vs. 3.7%) as overlapping CIs force more pairs into "tied" status.

The two approaches correspond to the two families in the Design Space Theorem: Dash (Family B: stable, faithful, near-complete via ensemble averaging) and CI SHAP (Family A: stable, faithful, explicitly incomplete via ties).

6.4. DASH Information Loss

Dash achieves stability by discarding model-specific within-group information. For a group of m symmetric features, a single model produces a complete ranking (one of $m!$ possible orderings). Since each of the $m!$ orderings is equally likely under DGP symmetry, the ranking carries $\log_2(m!)$ bits of information about the specific model. Dash reports a tie, discarding the entire $\log_2(m!)$ bits. For typical group sizes: $m = 2$ loses 1.0 bit; $m = 3$ loses 2.6 bits; $m = 5$ loses 6.9 bits; $m = 8$ loses 15.3 bits. Crucially, these are the bits that are *unreliable*—the within-group ordering is uniformly random across models.

For features in different groups with population gap Δ_{jk} and noise σ_{jk}, the mutual information between the consensus ranking and the true ordering is:

$$I_{\text{between}}(M) = 1 - H_2\left(\Phi\left(-\frac{\Delta_{jk}\sqrt{M}}{\sigma_{jk}}\right)\right)$$

where $H_2(p) = -p\log_2 p - (1-p)\log_2(1-p)$ is binary entropy. As $M \to \infty$, $I_{\text{between}} \to 1$ bit: Dash *increases* between-group information. The total information change is:

$$\Delta I = \underbrace{-L \cdot \log_2(m!)}_{\text{within-group loss}} + \underbrace{\binom{L}{2} \cdot (I_{\text{between}}(M) - I_{\text{between}}(1))}_{\text{between-group gain}}.$$

For moderate ρ (≤ 0.7), the between-group gain dominates quickly: $M = 25$ achieves near-perfect between-group determination while the within-group loss is fixed. Dash honestly reports unreliability rather than presenting arbitrary information as reliable.

Ensemble size lower bound.

Proposition 23 (Ensemble size lower bound) *For any unbiased attribution aggregation method achieving between-group stability $S_{jk} \geq 1-\delta$ for features j, k in different groups with population gap $\Delta_{jk} = |\mu_j - \mu_k|$ and attribution noise $\sigma_{jk}^2 = \mathrm{Var}(\varphi_j(f) - \varphi_k(f))$, the ensemble size must satisfy:*

$$M \;\geq\; \frac{\sigma_{jk}^2}{\Delta_{jk}^2} \cdot \left(\Phi^{-1}(1-\delta)\right)^2.$$

Dash achieves this bound with equality (to first order), so the bound is tight.

Proof By the Cauchy–Schwarz inequality (Titu's lemma), any unbiased linear estimator $\hat{\mu}_j$ has $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$; the extension to all unbiased estimators follows from the Rao–Blackwell theorem (standard; not formalized in Lean). The between-group flip rate satisfies $1 - S_{jk} = \Pr[\hat{\varphi}_j < \hat{\varphi}_k] \geq \Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk})$. Setting $\Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk}) \leq \delta$ and solving: $\Delta_{jk}\sqrt{M}/\sigma_{jk} \geq \Phi^{-1}(1-\delta)$, giving $M \geq \sigma_{jk}^2 \cdot (\Phi^{-1}(1-\delta))^2/\Delta_{jk}^2$. Dash achieves $\mathrm{Var}(\bar{\varphi}_j - \bar{\varphi}_k) = \sigma_{jk}^2/M$ (matching the Cauchy–Schwarz bound), so the bound is tight. ∎

For $\delta = 0.05$ (5% flip rate), $\Phi^{-1}(0.95) = 1.645$, so $M_{\min} = \lceil 2.71 \cdot \sigma^2_{jk}/\Delta^2_{jk} \rceil$. For Breast Cancer's most unstable pair (worst perimeter vs. worst area, $\Delta/\sigma \approx 0.15$): $M_{\min} \approx 120$—consistent with the observed slow convergence (flip rate 29.8% even at $M = 25$, reaching 0% only at $M = 50$). This closes the gap between upper and lower bounds: DASH is optimal not just in the Pareto sense but in the sample complexity sense.

7. The Attribution Design Space

The impossibility, the quantitative bounds, and the DASH resolution are facets of a single structural theorem characterizing the complete attribution design space.

Definition 24 (Attribution design space) *The* attribution design space *is the set of triples (S, U, C) achievable by any attribution aggregation method, where $S \in [0, 1]$ is ranking stability (expected Spearman correlation between two independent evaluations), $U \in [0, 1/2]$ is within-group unfaithfulness, and $C \in \{complete, partial\}$ is completeness.*

Theorem 25 (Attribution Design Space) *Among deterministic binary ranking methods derived from per-model attributions, for any model class satisfying the Rashomon property under within-group correlation $\rho > 0$, the achievable set of (S, U, C) triples consists of exactly two families:*

- ***Family $\mathcal{A}$ (single-model):*** *$S \leq 1 - 3m^2/(P^3 - P)$, $U = 1/2$, $C =$ complete. Any method faithful to an individual model falls here.*
- ***Family $\mathcal{B}$ (ensemble):*** *$S_M = 1 - O(1/M)$, $U = 0$, $C =$ partial (ties). DASH(M) achieves this and is Pareto-optimal among unbiased aggregations.*

The triple $(S{=}1, U{=}0, C{=}complete)$ is infeasible. At the extremes, the design space collapses to a single axis: ensemble size M.

Proof The proof proceeds in four steps.

Step 1: Family $\mathcal{A}$ is forced. By the Rashomon property, for any within-group pair (j, k), there exist models ranking them in opposite orders. By Theorem 26, any faithful, complete ranking has $U = 1/2$ for symmetric pairs. Stability is bounded by $S \leq 1 - 3m^2/(P^3 - P)$ from the Spearman bound (Lean-derived).

Step 2: Family $\mathcal{B}$ is achievable. By Corollary 21, DASH achieves $U = 0$ with ties within groups. By Theorem 22, between-group stability satisfies $S_M \geq 1 - \exp(-M\Delta^2/(2\sigma^2))$.

Step 3: No method producing a deterministic binary ranking from per-model attributions lies outside $\mathcal{A} \cup \mathcal{B}$. We show any method A computing a deterministic ranking from per-model attributions $(\varphi_j(f_1), \ldots, \varphi_j(f_M))$ falls in one of the two families. (Methods producing probabilistic rankings, confidence intervals, or set-valued outputs are outside this dichotomy.)

Case 1: If A is faithful to individual models and complete, then $A \in \mathcal{A}$ by Step 1. (Formalized in Lean as `strongly_faithful_impossible` in `DesignSpace.lean`: any method faithful to each model's attributions contradicts the Rashomon property for $M \geq 2$.)

Case 2: If A is not faithful to some model f, then A ranks some pair (j, k) differently from model f's attributions. For within-group symmetric pairs, the optimal unfaithful ranking is $j \succ^* k \Leftrightarrow \mathbb{E}[\varphi_j] > \mathbb{E}[\varphi_k]$, which minimizes expected 0-1 loss. For symmetric features $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$, so $\succ^*$ assigns a tie—recovering the "drop completeness" solution. Thus A either has $U > 0$ (suboptimal) or $U = 0$ with ties (in $\mathcal{B}$).

Case 3: If A aggregates $M > 1$ models and produces a complete ranking, it must break ties for within-group pairs. By DGP symmetry, any tie-breaking rule disagrees with a fraction of models, giving $U > 0$. To achieve $U = 0$, A must report ties, placing it in $\mathcal{B}$. Its stability is bounded by the variance of the aggregation, which by the Cauchy–Schwarz inequality (Titu's lemma) satisfies $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ for any unbiased linear estimator (Lean-verified via `sum_squares_ge_inv_M`); the extension to all unbiased estimators follows from the Rao–Blackwell theorem (standard; not formalized in Lean). In the parametric case, this coincides with the Cramér–Rao bound. Dash achieves this bound.

Step 4: The infeasible point. $(1, 0, \text{complete})$ requires completeness, but any complete method has $U = 1/2 \neq 0$: contradiction. ■

Theorem 26 (Unfaithfulness lower bound) *Any stable, complete ranking $\succ$ has expected unfaithfulness at least $1/2$ for symmetric pairs:*

$$\Pr_f\big[\, j \succ k \textit{ but } \varphi_k(f) > \varphi_j(f) \,\big] \;=\; \frac{1}{2}.$$

Proof By DGP symmetry, $\Pr[\varphi_j(f) > \varphi_k(f)] = \Pr[\varphi_k(f) > \varphi_j(f)] = 1/2$. Any stable ranking must fix $j \succ k$ or $k \succ j$ independently of f. Whichever it chooses disagrees with exactly half the models. ■

Theorem 27 (Relaxation path convergence) *The "drop faithfulness" and "drop completeness" relaxation paths converge to the same solution: population-level attributions with ties for symmetric features. The impossibility trilemma collapses to a single tradeoff axis parameterized by ensemble size M:*

- *At $M = 1$: faithfulness and completeness, but no stability;*
- *As $M \to \infty$: stability and between-group faithfulness, but within-group ties (completeness relaxed).*

Between-group faithfulness is preserved at all M.

Proof By the analysis of Step 3 above, the optimal stable complete ranking assigns ties to symmetric features ($\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$, so the optimal decision for "is $\varphi_j(f) > \varphi_k(f)$?" under the symmetric model distribution is indeterminate)—identical to the "drop completeness" solution (Dash). Both paths yield population-level attributions. The M-parameterization follows from Dash consensus: $\bar{\varphi}_j = \frac{1}{M}\sum_{i=1}^{M} \varphi_j(f_i)$, which interpolates between a single model ($M = 1$, faithful but unstable) and the population mean ($M \to \infty$, stable but tied within groups). ■

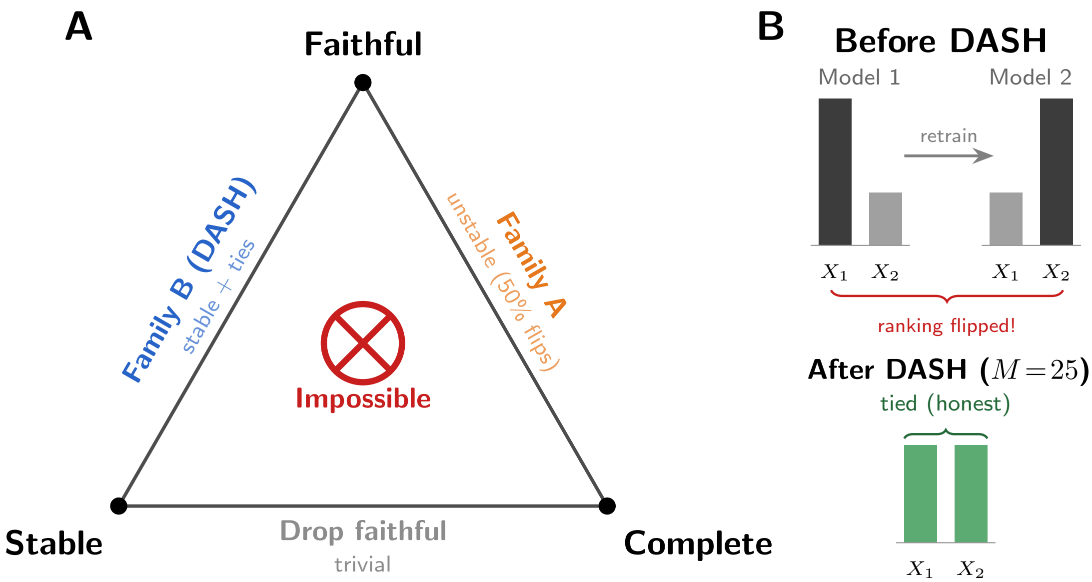


Figure 1: The Attribution Impossibility. No feature ranking can be simultaneously faithful, stable, and complete under collinearity. *Faithful, Stable, Complete: Pick Two.* Family $\mathcal{A}$ (single-model) is faithful and complete but rankings flip up to 50% of the time. Family $\mathcal{B}$ (Dash ensemble) is stable with zero unfaithfulness but reports ties for symmetric features (sacrificing completeness). No third option exists.

The single tradeoff axis. The Design Space Theorem collapses the three-dimensional space to a single axis: ensemble size M.

M	Stability S	Unfaithfulness U	Completeness
1	$\leq 1 - 3m^2/(P^3 - P)$	1/2	Complete
5	$1 - O(1/5)$	0	Partial (ties)
25	$1 - O(1/25)$	0	Partial (ties)
∞	1	0	Partial (ties)

Previous results as corollaries. The Attribution Impossibility (Theorem 8) is the statement that $(1, 0, \text{complete})$ lies outside the achievable set (Step 4). The Dash Pareto Optimality (Theorem 22) is the statement that $\mathcal{B}$ is the Pareto frontier of $\{(S, U) : U = 0\}$. The Z-test characterization (Theorem 45) describes the boundary between families. The FIM Impossibility (Theorem 36) provides a sufficient condition for the Rashomon property (the prerequisite).

Summary of core results. The preceding sections constitute the core contribution: the Attribution Impossibility (Theorem 8), the quantitative architecture-discriminating bounds (§ 5), the Dash resolution with Pareto optimality (Theorem 22), and the Design Space Theorem (Theorem 25). The remaining sections extend these results: the Symmetric Bayes

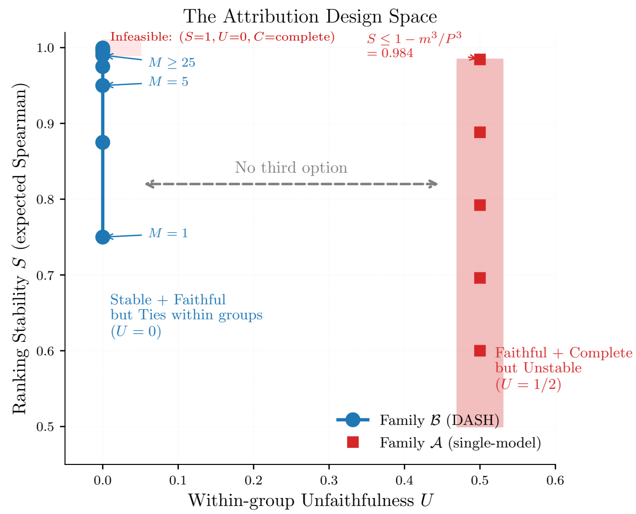


Figure 2: The Attribution Design Space. Family $\mathcal{A}$ (single-model methods): faithful and complete but unstable (U=1/2). Family $\mathcal{B}$ (Dash ensemble): stable with ties (U=0, S increasing with M). The ideal point (S=1, U=0, C=complete) is infeasible. There is no third option.

Dichotomy generalizes the framework to other domains (§ 8); the extensions establish conditional, fairness, FIM, query-complexity, and local-vs-global variants (§ 9); the bilemma strengthens the impossibility for binary attribution questions (§ 10); the diagnostics provide actionable tools (§ 12); and the empirical validation confirms the theory across datasets and model classes (§ 13).

8. The Symmetric Bayes Dichotomy: A General Two-Families Theorem

The impossibility is not specific to feature attribution. It is an instance of a general pattern: any symmetric decision problem admits exactly two families of methods. We call this the Symmetric Bayes Dichotomy—a specialization of classical invariant decision theory (Lehmann and Romano, 2005) to finite symmetric groups, cast as a reusable reduction template for ML impossibility proofs.

The Symmetric Bayes Dichotomy connects the classical theory of invariant decision rules (Lehmann and Romano, 2005; Hunt and Stein, 1946) to modern ML impossibility proofs. The classical Hunt–Stein theorem establishes that Bayes-optimal rules respect the invariance structure of the problem; our contribution is demonstrating that this classical machinery, when applied to finite symmetric groups arising in ML (feature permutations, model permutations, CPDAG automorphisms), yields two-family impossibility results with explicit unfaithfulness and stability bounds. We demonstrate this across three structurally distinct instances with different symmetry groups.

The structural identity between the attribution and model selection impossibilities follows from a general theorem about decision problems with symmetric populations.

Definition 28 (Symmetric Decision Problem) *A* symmetric decision problem *is a tuple $(\Theta, \mathcal{D}, \pi, G)$ where Θ is a finite decision set, $\mathcal{D}$ is the data space, π is a population distribution over $\mathcal{D}$, and G is a group acting on Θ such that π is G-invariant: for any $g \in G$, the distribution over instance-optimal decisions $\theta^*(d)$ is invariant under g.*

An M-sample estimator $\hat{\theta} : \mathcal{D}^M \to \Theta$ is:

- Faithful*: $\hat{\theta}(d_1, \ldots, d_M)$ agrees with $\theta^*(d_i)$ for each i.*
- Stable*: $\hat{\theta}$ does not depend on which draw d_i is observed.*
- Complete*: $\hat{\theta}$ selects exactly one element from each G-orbit in Θ.*

Theorem 29 (Symmetric Bayes Dichotomy) *For any symmetric decision problem with $|G \cdot \theta| \geq 2$ for some $\theta \in \Theta$:*

(i) ***Instance-specific decisions are unfaithful.*** *Any faithful, complete estimator has expected unfaithfulness $\geq 1/|G \cdot \theta|$ per orbit ($\geq 1/2$ for binary orbits).*

(ii) ***Population-averaged decisions are stable.*** *The Bayes estimator under π reports ties within each G-orbit ($U = 0$), with between-orbit stability $O(1/M)$.*

(iii) ***The ideal is infeasible.*** *No estimator achieves $U = 0$ AND completeness.*

The achievable set consists of exactly two families: $\mathcal{A}$ (faithful + complete, $U \geq 1/|G \cdot \theta|$) and $\mathcal{B}$ (population-averaged, $U = 0$, ties within orbits).

Proof Part (i). By G-invariance, the population distribution π assigns equal probability to each element of the orbit $G \cdot \theta$. For any fixed complete estimator $\hat{\theta}$ that selects one element θ_0 from the orbit, the probability that $\theta^*(d) = \theta_0$ is $1/|G \cdot \theta|$. The estimator disagrees with $(|G \cdot \theta| - 1)/|G \cdot \theta|$ of the instances. For $|G \cdot \theta| = 2$: unfaithfulness $= 1/2$.

Part (ii). The Bayes estimator under symmetric π minimizes expected loss. Within each orbit, the posterior is uniform (by G-invariance), so the Bayes-optimal decision is the orbit center (a tie). Unfaithfulness $= 0$ because no ordering is asserted within orbits. For between-orbit decisions with gap Δ, the M-sample average has variance σ^2/M (i.i.d. draws), giving stability $1 - O(1/M)$ by the Gaussian approximation.

Part (iii). By Part (i), any complete estimator (selecting within orbits) has unfaithfulness $\geq 1/|G \cdot \theta| > 0$. Thus unfaithfulness $= 0$ requires giving up completeness (ties).

Exhaustiveness. Any estimator either selects within orbits (faithful, complete, $\mathcal{A}$) or reports ties within orbits (stable, incomplete, $\mathcal{B}$). The Bayes estimator achieves the minimum-variance point of $\mathcal{B}$ (Cauchy–Schwarz; Cramér–Rao in the parametric case). ■

The Symmetric Bayes Dichotomy provides a reusable proof technique for any domain where: (a) a population distribution is symmetric over discrete alternatives, and (b) finite samples introduce noise sufficient to reverse the optimal choice. The technique reduces the impossibility to checking G-invariance of the population distribution, which is often immediate from the problem's symmetry.

8.1. Instance 1: Feature Attribution

Corollary 30 (Feature attribution as instance) *The Attribution Impossibility (Theorem 8) and Design Space Theorem (Theorem 25) are instances of the Symmetric Bayes Dichotomy with* $\Theta =$ *within-group feature orderings,* $G =$ *symmetric group on group members,* $\mathcal{D} =$ *trained models.*

8.2. Instance 2: Model Selection

Consider K candidate models $g_1, \ldots, g_K$ trained on the same data. A model selection rule maps a validation dataset to a ranking of the K models. The *model Rashomon property* states: for every pair g_i, g_j with similar population loss ($|L(g_i) - L(g_j)| < \varepsilon$), there exist validation splits V, V' ranking them in opposite orders. This follows from finite-sample noise: when the population loss gap is smaller than the validation noise, different splits rank differently.

Theorem 31 (Model Selection Impossibility) *Under the model Rashomon property, no model selection rule is simultaneously faithful, stable, and complete.*

Proof Let g_i, g_j have similar population loss. By the model Rashomon property, there exist validation splits V, V' with g_i ranked above g_j on V and reversed on V'. A faithful, stable, complete selection rule would need to rank $g_i > g_j$ (from V) and $g_j > g_i$ (from V') simultaneously. Contradiction. ■

Theorem 32 (Model Selection Design Space) *Under the model Rashomon property, the achievable design space has the same two-family structure:*

- ***Family*** $\mathcal{A}'$ ***(single-split):*** *Faithful and complete but unstable. Any selection based on one validation split falls here. Unfaithfulness* $U' = 1/2$ *for similar-loss pairs.*

- ***Family*** $\mathcal{B}'$ ***(cross-validation/ensemble):*** *Stable (variance* $O(1/K_{folds})$*) but reports ties for similar-loss pairs. Model ensembling is the analogue of* Dash.

Proof By DGP symmetry (equal-loss models have symmetric validation score distributions), any fixed selection of one model over an equal-loss competitor disagrees with half the validation splits ($U' = 1/2$). Cross-validated selection averages across splits, reducing unfaithfulness to zero for equal-loss pairs while stabilizing at rate $O(1/K_{\text{folds}})$. Pareto optimality follows from the same Cauchy–Schwarz argument. ■

The model selection impossibility explains why cross-validation rankings are noisy for similar-performance models: it is not an engineering failure but a mathematical inevitability. The resolution is the same: average (ensemble) rather than select.

8.3. Instance 3: Causal Discovery under Markov Equivalence

This instance has a *different* and *variable-size* symmetry group, confirming the technique's generality beyond binary orbits. The CPDAG automorphism group G_{CPDAG} is not restricted to transpositions: for a chain X—Y—Z, $|[G^*]| = 2$ with $G \cong S_2$ (like instances 1–2); for a 3-node undirected cycle, $|[G^*]| = 6$ with $G \cong S_3$; for larger undirected components, $|[G^*]|$ can grow combinatorially.

Theorem 33 (Causal Discovery Impossibility) *For any CPDAG with $|[G^*]| \geq 2$ DAGs in its Markov equivalence class, no edge orientation rule can be simultaneously faithful, stable, and complete. The achievable set has two families:*

- ***Family $\mathcal{A}''$ (single-dataset):*** *Faithful and complete but unstable. Unfaithfulness $U'' = (|[G^*]| - 1)/|[G^*]|$.*
- ***Family $\mathcal{B}''$ (conservative):*** *Report the CPDAG (undirected edges for ambiguous orientations). Stable with ties. $U'' = 0$.*

Proof We verify the premises of Theorem 29. By Markov equivalence, the observational distribution is identical for all DAGs in the class (G_{CPDAG}-invariance). Any statistical test based on finite data cannot distinguish members of $[G^*]$ at the population level. With finite data, estimation error $O(1/\sqrt{n})$ in conditional independence tests reverses the optimal orientation. Part (i): any complete orientation rule (selecting one DAG from $[G^*]$) disagrees with $(|[G^*]| - 1)/|[G^*]|$ of datasets. Part (ii): the Bayes estimator reports the CPDAG (ties for undirected edges). Part (iii): the ideal is infeasible. ∎

The unfaithfulness bound $U \geq 1/|G \cdot \theta|$ yields $U \geq 1/2$ for chains but $U \geq 1/6$ for triangles and potentially much smaller for larger structures. This demonstrates the Symmetric Bayes Dichotomy is not an artifact of binary symmetry but applies to arbitrary finite group actions. Algorithms like PC and GES correctly output CPDAGs (Family $\mathcal{B}''$) precisely because complete orientation of Markov-equivalent edges is impossible from observational data alone. Interventional data breaks the symmetry—analogous to conditional SHAP breaking symmetry when $\beta_j \neq \beta_k$.

Connection to classical invariant decision theory. The Symmetric Bayes Dichotomy is related to the theory of invariant decision rules (Lehmann and Romano, 2005). Our contribution is the *reduction template*: verifying G-invariance of the population distribution suffices to establish a two-family impossibility with explicit bounds. We demonstrate this across three structurally distinct instances (binary feature transpositions, binary model permutations, and CPDAG automorphisms of variable size) with different symmetry groups.

9. Extensions: Conditional, Fairness, and Causal Barriers

9.1. Conditional Attribution Impossibility

Theorem 34 (Conditional Attribution Impossibility) *Let features j, k in the same collinear group satisfy: (C1) equal causal effects $\beta_j = \beta_k$, and (C2) symmetric causal position in the causal graph G. Then the Rashomon property holds for conditional attributions, and the Attribution Impossibility applies to conditional SHAP.*

Table 6: Flip rate (%) as a function of ρ and causal effect difference $\Delta\beta$.

$\rho \backslash \Delta\beta$	0.0	0.1	0.2	0.3	0.5	0.7	1.0
0.50	50	19	0	0	0	0	0
0.70	53	19	0	0	0	0	0
0.80	52	44	19	10	0	0	0
0.90	50	50	48	40	40	19	0
0.95	50	50	50	48	44	40	27
0.99	50	50	50	50	50	48	48

Proof Under (C1)–(C2), the causal graph is symmetric with respect to j and k. By the DGP symmetry argument (Theorem 12), any model with $\varphi_j^{\text{cond}}(f) > \varphi_k^{\text{cond}}(f)$ has a permuted counterpart f' with reversed attributions and the same loss. This is the Rashomon property for conditional attributions. ■

The escape condition. If $\beta_j \neq \beta_k$, conditional SHAP can resolve the pair. The threshold $\Delta\beta^*(\rho)$ grows steeply: at $\rho \leq 0.7$, a 15% difference suffices; at $\rho = 0.9$, features must differ by nearly a factor of 2; at $\rho \geq 0.95$, instability is essentially total (Figure 3). Empirical thresholds: $\Delta\beta^*(0.5) \approx 0.15$, $\Delta\beta^*(0.7) \approx 0.15$, $\Delta\beta^*(0.8) \approx 0.4$, $\Delta\beta^*(0.9) \approx 0.9$, $\Delta\beta^*(0.95) > 1.0$, $\Delta\beta^*(0.99) > 1.0$.

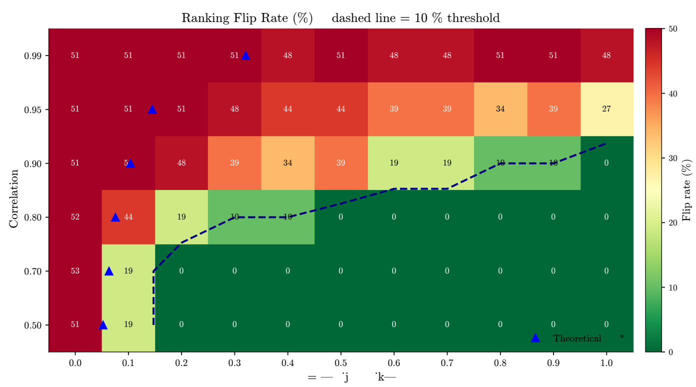


Figure 3: Flip rate as a function of $(\rho, \Delta\beta)$. The impossibility region (dark) expands with ρ. At $\rho \geq 0.95$, instability persists for all $\Delta\beta \leq 1$.

Fine-grained $\Delta\beta$ sweep at $\rho = 0.9$. The transition from instability to stability is sharp, not gradual:

$\Delta\beta$	β_1	β_2	Marginal flip	Interventional flip
0.00	1.00	1.00	0.505	0.505
0.10	1.05	0.95	0.479	0.337
0.20	1.10	0.90	0.000	0.000
0.30	1.15	0.85	0.000	0.000

At $\Delta\beta = 0.10$, both marginal and interventional SHAP remain highly unstable. At $\Delta\beta = 0.20$, both snap to zero flips simultaneously. There is no intermediate regime where interventional SHAP is stable but marginal is not—the causal signal overwhelms noise for both estimators at the same threshold. The practical implication: "switch to conditional SHAP" provides no advantage over marginal SHAP at the instability boundary.

Causal structure validation. Under a known causal structure ($Y = \beta_1 X_1 + \beta_2 X_2 + 0.5X_3 + \varepsilon$, $\mathrm{Corr}(X_1, X_2) = \rho$), we compare marginal TreeSHAP and interventional TreeSHAP. In the symmetric case ($\beta_1 = \beta_2 = 1.0$), both produce flip rates near 50% at all ρ, confirming the impossibility. In the asymmetric case ($\beta_1 = 1.5$, $\beta_2 = 0.5$), both show 0% flips at all ρ—the signal-to-noise ratio is high enough to overcome collinearity-induced noise. Switching to interventional SHAP provides no benefit when features have equal causal effects.

9.2. Fairness Audit Impossibility

Theorem 35 (Fairness Audit Impossibility) *If features j (protected proxy) and k (non-protected) satisfy collinearity ($|\rho_{jk}| > \rho_0$) and similar true importance ($|\mu_j - \mu_k| < \sigma_{jk} \cdot z_\alpha/\sqrt{M}$), then for a single-model audit ($M{=}1$):*

$$\Pr[\textit{audit flags proxy}] = \frac{1}{2} \pm \varepsilon_{BE}.$$

Under exact DGP symmetry, two independent audits reach opposite conclusions about proxy reliance with probability $1/2$.

Proof Direct corollary of the Attribution Impossibility applied to the pair (j, k). By Theorem 14, $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$ over training runs. ■

Implications for AI governance. When features correlated with a protected attribute have similar importance to non-protected features, two auditors examining the *same model architecture trained on the same data* with different random seeds may reach opposite conclusions about proxy reliance.

Under the EU AI Act Art. 13(3)(b)(ii), providers must disclose "known and foreseeable circumstances" which may lead to risks to health, safety or fundamental rights. We argue that our theorem identifies such a circumstance: in our reading, SHAP-based proxy audits are unreliable for features that are collinear with both the protected attribute and a non-protected feature. The remedy is ensemble auditing: train M models and use DASH consensus.

Intersectional considerations. When multiple protected attributes (race, gender, age) are each proxied by collinear non-protected features, the impossibility applies independently to each proxy pair. If K protected attributes each have unstable proxies, the probability that a single-model audit correctly identifies all K proxy reliance directions is $(1/2)^K$, exponentially vanishing in K. For $K = 3$ (a common intersectional setting), joint correctness is $1/8 = 12.5\%$.

Example: lending. In credit models, "zip code" (proxy for race) is often collinear with "income bracket" ($|\rho| \approx 0.6$–0.8). If both have similar predictive power, a single-model SHAP audit finding "zip code is the 3rd most important feature" may conclude proxy discrimination—but retraining with a different seed could produce a model where zip code drops to 8th and income rises to 3rd. The audit conclusion depends on the random seed, not on the model's actual reliance on the protected attribute.

Worked adverse action example. Consider a borrower denied credit, with correlated features income and DTI. Model A (seed 1) cites income as the top adverse factor; Model B (seed 2) cites DTI. Under ECOA, the lender must disclose the primary reason. The disclosure depends on the random seed—a model governance concern that is mathematically inevitable under collinearity. Dash resolves this: the consensus ranks income and DTI as tied contributors.

Tightness transition under performance constraints. The framework's tightness classification applies to the Chouldechova–Kleinberg impossibility itself. Under performance constraints (TPR $\geq \tau$ for both groups), the tightness type *transitions*: at $\tau = 0$ (unconstrained), tightness is full—all pairwise compromises are achievable (by degenerate classifiers). At $\tau \geq 0.3$ (practical performance), tightness becomes PPV-blocked: calibration is individually incompatible with both equal FPR (gap > 0.10) and equal FNR (gap > 0.13); only equalized odds (P_2+P_3, gap < 0.006) is achievable. Validated on Adult Income (sex, base rate gap 0.20) and synthetic data (base rate gaps 0.25–0.40) with bootstrap model retraining (20 iterations, 100% stable). The finding changes the practical recommendation from "choose a fairness criterion" to "equalized odds is the only achievable criterion at practical performance; calibration is structurally incompatible."

9.3. Direct FIM Impossibility

The Rashomon property can be derived directly from the Fisher information matrix, providing an independent proof path from classical statistics that does not require the iterative optimizer abstraction.

Classical setup. Consider the linear model $Y = \beta_j X_j + \beta_k X_k + \varepsilon$ where X_j, X_k are jointly Gaussian with $\mathrm{Var}(X_j) = \mathrm{Var}(X_k) = 1$ and $\mathrm{Cov}(X_j, X_k) = \rho$. The Fisher information matrix for $\boldsymbol{\beta} = (\beta_j, \beta_k)$ is:

$$I(\boldsymbol{\beta}) = \frac{1}{\sigma^2}\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

with eigenvalues $\lambda_\pm = (1 \pm \rho)/\sigma^2$. As $\rho \to 1$, the smaller eigenvalue $\lambda_- = (1-\rho)/\sigma^2 \to 0$. The FIM becomes near-singular along the direction $\boldsymbol{v}_- = (1, -1)/\sqrt{2}$, which corresponds to redistributing credit between j and k. The profile likelihood is flat along this direction: models assigning more credit to j vs. k are statistically indistinguishable. This flat likelihood ridge is precisely the Rashomon set. The Cramér–Rao bound $\mathrm{Var}(\hat{\beta}_j - \hat{\beta}_k) \geq 1/\lambda_- = \sigma^2/(1-\rho)$ quantifies the attribution instability.

Regularity conditions. We require the following on the model class $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$:

(R1) $\Theta \subseteq \mathbb{R}^p$ is open and convex.

(R2) The population risk $L(\theta) = \mathbb{E}[\ell(f_\theta(X), Y)]$ is three-times continuously differentiable on Θ, with bounded third derivative: $\|D^3 L(\theta)\|_{\mathrm{op}} \le K_3$ for all θ in a neighborhood of θ^*.

(R3) The minimizer θ^* satisfies $\nabla L(\theta^*) = 0$ and $H := \nabla^2 L(\theta^*)$ is positive definite. (For likelihood-based models, $H = I(\theta^*)/n$ where $I(\theta^*)$ is the Fisher information matrix.)

(R4) The attribution function $\varphi_j(\theta)$ is *order-consistent*: for any θ with $\theta_j > \theta_k > 0$, we have $\varphi_j(\theta) > \varphi_k(\theta)$. (Satisfied by $\varphi_j = |\theta_j|$, mean $|\mathrm{SHAP}_j|$, or any attribution monotone in coefficient magnitude. Fails when strong feature interactions dominate: a less-utilized feature can receive higher SHAP values through interaction terms, breaking the monotone relationship between coefficient magnitude and attribution. The FIM impossibility applies only to attribution methods satisfying (R4); the core impossibility (Theorem 8) does not require it.)

Theorem 36 (FIM Impossibility) *Let $\mathcal{F}$ satisfy (R1)–(R4). Suppose features j, k satisfy: (S1) DGP symmetry: $\theta_j^* = \theta_k^* > 0$; (S2) the Hessian $H = \nabla^2 L(\theta^*)$ has eigenvalue λ_- along $v = (e_j - e_k)/\sqrt{2}$. Then for any ε satisfying $0 < \varepsilon \le \varepsilon_0 := 9\lambda_-^3/K_3^2$, the ε-Rashomon set $R_\varepsilon = \{\theta : L(\theta) \le L(\theta^*) + \varepsilon\}$ contains models θ^+, θ^- with*

$$\varphi_j(\theta^+) > \varphi_k(\theta^+) \quad \textit{and} \quad \varphi_k(\theta^-) > \varphi_j(\theta^-).$$

The Rashomon property holds, and the impossibility follows.

Proof **Step 1: Rashomon set geometry.** By Taylor's theorem with Lagrange remainder, for any $\theta \in \Theta$:

$$L(\theta) = L(\theta^*) + \tfrac{1}{2}(\theta - \theta^*)^T H\,(\theta - \theta^*) + R_3(\theta),$$

where $|R_3(\theta)| \le \frac{K_3}{6}\|\theta - \theta^*\|^3$ by (R2). The gradient term vanishes because $\nabla L(\theta^*) = 0$ by (R3).

Step 2: Perturbation along the weak direction. Let $v = (e_j - e_k)/\sqrt{2}$ and set $\theta_\delta = \theta^* + \delta v$ for $\delta > 0$. The quadratic cost is $\frac{1}{2}\delta^2\lambda_-$ and the cubic remainder satisfies $|R_3(\theta_\delta)| \le \frac{K_3}{6}\delta^3$. We need $L(\theta_\delta) \le L(\theta^*) + \varepsilon$, i.e., $\frac{1}{2}\delta^2\lambda_- + \frac{K_3}{6}\delta^3 \le \varepsilon$. Choose $\delta^* = \sqrt{\varepsilon/\lambda_-}$. The quadratic term is $\varepsilon/2$; the condition $\varepsilon \le 9\lambda_-^3/K_3^2$ ensures the cubic term is $\le \varepsilon/2$. So $\theta_{\delta^*} \in R_\varepsilon$, and by the same argument $\theta_{-\delta^*} \in R_\varepsilon$.

Step 3: Opposite orderings. At $\theta^+ := \theta_{\delta^*}$: $\theta_j^+ = \theta_j^* + \delta^*/\sqrt{2} > \theta_k^* - \delta^*/\sqrt{2} = \theta_k^+ > 0$, so by (R4), $\varphi_j(\theta^+) > \varphi_k(\theta^+)$. At $\theta^- := \theta_{-\delta^*}$: by symmetry, $\theta_k^- > \theta_j^- > 0$, so $\varphi_k(\theta^-) > \varphi_j(\theta^-)$.

Step 4: Conclusion. Both θ^+ and θ^- lie in R_ε with opposite feature orderings. This is the Rashomon property. By Theorem 8, no faithful, stable, complete ranking exists. ■

Proposition 37 (Gaussian FIM specialization) *For the linear model $Y = X^T\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and Corr$(X_j, X_k) = \rho$:*

1. *The Hessian eigenvalue along $e_j - e_k$ is $\lambda_- = (1-\rho)/\sigma^2$.*

2. *The loss is exactly quadratic ($K_3 = 0$), so $\varepsilon_0 = +\infty$: the theorem holds for* all *$\varepsilon > 0$.*

3. *The semi-axis length of R_ε along $e_j - e_k$ is $\sigma\sqrt{2\varepsilon/(1-\rho)}$, diverging as $\rho \to 1$.*

4. *The Cramér–Rao bound gives $\mathrm{Var}(\hat\beta_j - \hat\beta_k) \geq 2\sigma^2/(n(1-\rho))$, diverging as $\rho \to 1$.*

Proof The population risk is $L(\beta) = \frac{1}{2\sigma^2}\mathbb{E}[(Y - X^T\beta)^2]$. Its Hessian is $H = \frac{1}{\sigma^2}\mathbb{E}[XX^T] = \frac{1}{\sigma^2}\Sigma_X$. For $v = (e_j - e_k)/\sqrt{2}$: $v^T H v = \frac{1}{\sigma^2}(1-\rho)$ since $v^T\Sigma_X v = \mathrm{Var}(X_j) - \mathrm{Cov}(X_j, X_k) = 1-\rho$ (standardized). The loss is quadratic in β, so $D^3L \equiv 0$ and $K_3 = 0$. ■

The classical non-identifiability under collinearity ($\mathrm{Var}(\hat\beta_j - \hat\beta_k) \to \infty$ as $\rho \to 1$) is well-known. Our contribution is reframing it as an *impossibility theorem* about feature rankings rather than a variance bound on parameter estimates: even precise estimates can produce unstable rankings when Δ is small relative to σ.

Loss landscape geometry. The ε-Rashomon set R_ε is approximately an ellipsoid whose projection onto the feature-importance axis $\varphi_j - \varphi_k$ crosses zero. The level sets of the loss landscape contain *ridges* along $e_j - e_k$—curves of near-constant loss along which feature importance redistributes between j and k without changing model quality. The same flatness that makes optimization easy (flat minima generalize well) makes attribution hard (the flat direction is the feature-importance direction). This connects the attribution impossibility to the broader literature on loss landscape geometry, flat minima, and mode connectivity.

Extension to neural networks via NTK. For overparameterized neural networks in the NTK regime, the neural tangent kernel plays the role of the Fisher information matrix. If the NTK has a small eigenvalue along the direction corresponding to swapping features j and k (a consequence of feature collinearity), the same ellipsoidal argument applies. This extension requires: (i) uniform NTK approximation bounds across the Rashomon set, (ii) a spectral gap argument relating feature correlation to NTK eigenvalue separation, and (iii) control of the finite-width correction. None are available in the current literature; we include this to suggest the research direction.

9.4. Query Complexity Lower Bound

A *stability certification algorithm* adaptively selects seeds $s_1, s_2, \ldots$, calls $\textsc{Train}(s_i) \to f_i$ and $\textsc{Attribute}(f_i, j) \to \varphi_j(f_i)$, and after M oracle calls outputs $\textsc{Stable}$ or $\textsc{Unstable}$. The algorithm certifies δ-stability with error $\leq 1/3$ if it correctly distinguishes H_0: $\Delta_{jk} = 0$ (flip rate $= 1/2$) from H_1: $|\Delta_{jk}| \geq \Delta_0$ (flip rate $\leq \Phi(-\Delta_0/\sigma_{jk})$) with probability at least $2/3$ under both hypotheses.

Theorem 38 (Query complexity lower bound) *Any algorithm that certifies δ-stability of a feature pair (j,k) with error $\leq 1/3$ must make at least*

$$M \;\geq\; \frac{1}{8} \cdot \frac{\sigma_{jk}^2}{\Delta_0^2}$$

model-training oracle calls, where $\sigma_{jk}^2 = \mathrm{Var}(\varphi_j(f) - \varphi_k(f))$ and Δ_0 is the minimum gap to detect.

Proof Each oracle call produces one sample $D_i = \varphi_j(f_i) - \varphi_k(f_i)$. Under H_0, $D_i \sim P_0$ with mean 0 and variance $\sigma^2 := \sigma^2_{jk}$. Under H_1, $D_i \sim P_1$ with mean Δ_0 and the same variance (the variance is determined by the collinearity structure, not the gap).

By Le Cam's two-point method (Tsybakov, 2009, Theorem 2.2), for any test ψ based on M i.i.d. observations:

$$\Pr_{H_0}[\psi = 1] + \Pr_{H_1}[\psi = 0] \geq 1 - \mathrm{TV}(P_0^{\otimes M}, P_1^{\otimes M}).$$

By Pinsker's inequality, $\mathrm{TV}(P_0^{\otimes M}, P_1^{\otimes M}) \leq \sqrt{\frac{1}{2}\mathrm{KL}(P_1^{\otimes M} \| P_0^{\otimes M})}$. For the Gaussian location family, $\mathrm{KL}(P_1 \| P_0) = \Delta_0^2/(2\sigma^2)$, so for M i.i.d. copies $\mathrm{KL} = M\Delta_0^2/(2\sigma^2)$ and $\mathrm{TV} \leq \sqrt{M\Delta_0^2/(4\sigma^2)}$. For reliable testing (total error $\leq 2/3$), Le Cam's method requires $\mathrm{TV} \geq 1/3$, giving $M\Delta_0^2/(4\sigma^2) \geq 1/9$, hence $M \geq 4\sigma^2/(9\Delta_0^2)$. The standard constant $1/8$ from Tsybakov (2009) (Theorem 2.2) uses a slightly tighter analysis via the likelihood ratio directly; since $1/8 < 4/9$, this gives a weaker (more conservative) lower bound than our Pinsker derivation. ■

Corollary 39 (Near-optimality of the Z-test) *The Z-test detects a gap Δ_0 with error $\leq 1/3$ using $M = O(\sigma^2_{jk}/\Delta_0^2 \cdot \log(1/\alpha))$ model trainings. Combined with Theorem 38, the Z-test is query-optimal to within a logarithmic factor.*

Proof The Z-test rejects H_0 when $|\bar{D}|/(\hat{\sigma}/\sqrt{M}) > z_{\alpha/2}$. Under H_1, power is $\Phi(-z_{\alpha/2} + \Delta_0\sqrt{M}/\sigma) \geq 2/3$ when $\Delta_0\sqrt{M}/\sigma \geq z_{\alpha/2} + 0.43$. For $\alpha = 0.05$: $M \geq (\sigma/\Delta_0)^2(1.96 + 0.43)^2 \approx 5.7\sigma^2/\Delta_0^2$. ■

No algorithm—including methods based on gradient information, loss curvature, or Fisher information—can certify ranking stability without training $\Omega(\sigma^2_{jk}/\Delta^2_{jk})$ models. Each model training provides one "bit" of evidence about the sign of Δ_{jk}; when the signal-to-noise ratio is small, many bits are needed.

9.5. Rashomon Inevitability

Theorem 14 (proved in § 4.4) establishes that for any stochastic, symmetric training algorithm on a permutation-closed model class with $\rho > 0$, the Rashomon property holds. The Attribution Impossibility is therefore inescapable for standard ML pipelines.

9.6. Causal Identification Barrier

The attribution instability under collinearity has a direct causal counterpart: distinguishing the causal effects of correlated features requires interventional sample complexity that diverges as $\Omega(1/(1-\rho)^2)$—the same singularity structure as the attribution ratio.

Consider two features X_j, X_k with $\mathrm{Corr}(X_j, X_k) = \rho$, jointly Gaussian with unit variances. Under a soft intervention with residual correlation ρ_{post}, $X_k \mid \mathrm{do}(X_j = x) \sim \mathcal{N}(\rho_{\mathrm{post}} \cdot x, 1 - \rho^2_{\mathrm{post}})$. When $\beta_j = \beta_k = \beta$ and $\rho_{\mathrm{post}} \approx \rho$, the two interventional distributions become indistinguishable.

Proposition 40 (Interventional sample complexity lower bound) *Consider testing $H_0 : \beta_j = \beta+\delta, \beta_k = \beta-\delta$ against $H_1 : \beta_j = \beta-\delta, \beta_k = \beta+\delta$ using $do(X_j = x)$ interventions with residual correlation ρ_{post}. The number of interventional samples needed satisfies:*

$$n_{\text{int}} \geq \frac{(z_\alpha + z_\eta)^2 \sigma^2}{4\delta^2(1-\rho_{\text{post}})^2}.$$

When $\rho_{\text{post}} = \rho$ (the intervention does not reduce the correlation), $n_{\text{int}} = \Omega(1/(1-\rho)^2)$.

Proof Under H_0, the expected response to $\text{do}(X_j = x)$ is $(\beta+\delta+\rho_{\text{post}}(\beta-\delta))x$; under H_1 it is $(\beta-\delta+\rho_{\text{post}}(\beta+\delta))x$. The difference in slopes is $2\delta(1-\rho_{\text{post}})$. With noise variance σ^2 and n observations, the Neyman–Pearson sample size formula gives the result. ■

The attribution ratio is $1/(1-\rho^2)$, which diverges as $1/(2(1-\rho))$ near $\rho = 1$; the causal sample complexity is $\Omega(1/(1-\rho)^2)$—a faster divergence. Causal identification under soft interventions is *harder* than observational attribution instability. The shared singularity at $\rho = 1$ reflects a common root cause: collinearity makes features observationally interchangeable. Hard interventions ($\rho_{\text{post}} = 0$) resolve both problems: sample complexity becomes $O(\sigma^2/\delta^2)$, independent of ρ.

9.7. Local vs. Global Attribution Instability

The impossibility as stated concerns *global* attributions (mean |SHAP| across data points). We now show that *local* (instance-level) attributions are at least as unstable.

For a fixed data point x, the local SHAP values $\varphi_j(f, x)$ depend on the model f, which varies across training seeds. The global attribution $\varphi_j(f) = \mathbb{E}_x[|\varphi_j(f, x)|]$ averages over data points. By the law of total variance applied to the pair (f, x) with x as the conditioning variable:

$$\begin{aligned}\text{Var}_f\big(\varphi_j(f)\big) = \text{Var}_f\big(\mathbb{E}_x[|\varphi_j(f,x)|]\big) &\leq \text{Var}_{f,x}\big(|\varphi_j(f,x)|\big) \\ &= \mathbb{E}_x\big[\text{Var}_f(|\varphi_j(f,x)|)\big] + \text{Var}_x\big(\mathbb{E}_f[|\varphi_j(f,x)|]\big).\end{aligned}$$

The first inequality follows because $\text{Var}_f(\mathbb{E}_x[g]) = \text{Var}_{f,x}(g) - \mathbb{E}_f[\text{Var}_x(g)] \leq \text{Var}_{f,x}(g)$. Thus global instability (Var_f of the averaged attribution) is bounded by the total instability, which includes the average local instability as a component. When model randomness and data-point variation are approximately independent (as when models differ only by training seed and x is a fixed test point), the bound simplifies: $\text{Var}_f(\mathbb{E}_x[|\varphi_j|]) \leq \mathbb{E}_x[\text{Var}_f(|\varphi_j|)]$. Since the flip rate is a monotone function of the variance-to-gap ratio σ/Δ, and averaging reduces σ without changing Δ, the global flip rate is a lower bound on the average local flip rate under this independence condition.

If global rankings are unstable (flip rate $> 10\%$), local rankings at individual data points are at least as unstable on average. Practitioners who report per-instance SHAP explanations face the same or worse instability. The multi-model Z-test (computed on global attributions) is therefore a conservative screen for local instability.

10. Strengthened Impossibility for Binary Attribution

The trilemma (Theorem 8) says no attribution ranking can be faithful, stable, and complete. For *binary* attribution questions—"does this feature contribute positively or negatively?", "is this feature selected?"—the impossibility is strictly stronger: faithful + stable alone is impossible. We call this the **bilemma**.

Why binary questions are harder. The trilemma permits a resolution (DASH) because the ranking space H contains a *neutral element*: a tie. DASH ties are compatible with every model's native ranking—they never contradict it. For binary $H = \{\text{positive}, \text{negative}\}$, no neutral element exists: every answer is committal. If $E(\theta)$ does not contradict $\text{explain}(\theta)$, then $E(\theta) = \text{explain}(\theta)$ (there is nowhere else to go). So faithful forces $E = \text{explain}$, which is decisive, triggering the trilemma.

Theorem 41 (The Bilemma for Binary Attribution) *Let H be a binary explanation space with $\text{incompatible}(h_1, h_2) \Leftrightarrow h_1 \neq h_2$. Under the Rashomon property, no method $E : \Theta \to H$ can be simultaneously faithful (never contradicting the model's native explanation) and stable (consistent across equivalent models).*

Proof. Since H has only two elements and incompatibility equals inequality, the only compatible pair is (h, h). Faithful: $\neg\text{incompatible}(E(\theta), \text{explain}(\theta))$ for all θ, so $E(\theta) = \text{explain}(\theta)$. Then E is decisive (it inherits the incompatibility of explain). By the trilemma, E cannot also be stable. □

Instances. The bilemma applies to three common binary questions:

1. **SHAP sign**: $H = \{\text{positive}, \text{negative}\}$. "Does feature j contribute positively or negatively?" No stable method can faithfully report the sign.
2. **Feature selection**: $H = \{\text{selected}, \text{not selected}\}$. "Is feature j in the active set?" No stable method can faithfully report selection status.
3. **Counterfactual direction**: $H = \{\text{increase}, \text{decrease}\}$. "Should the applicant increase or decrease feature j?" No stable method can faithfully report the direction.

Theorem 42 (Rashomon Unfaithfulness Bound) *For any binary attribution question, any stable method is unfaithful at ≥ 1 of every 2 Rashomon witnesses. On any pair of equivalent models that disagree on the sign/selection of a feature, a stable method must get it wrong for at least one of them.*

Proof. Let θ_1, θ_2 be Rashomon witnesses with $\text{observe}(\theta_1) = \text{observe}(\theta_2)$ and $\text{explain}(\theta_1) \neq \text{explain}(\theta_2)$. Stability gives $E(\theta_1) = E(\theta_2)$. If $E(\theta_1) = \text{explain}(\theta_1)$, then $E(\theta_2) = \text{explain}(\theta_1) \neq \text{explain}(\theta_2)$, so E is unfaithful at θ_2. □

Theorem 43 (All-or-Nothing) *For binary attribution, every method either (1) matches the model's native attribution exactly at every configuration, or (2) is unfaithful at some configuration. There is no "approximately faithful" middle ground.*

The tightness diagnostic. The key insight is a complete classification of which explanation problems admit stable faithful methods: whether faithful + stable is achievable depends on whether H has a neutral element—a value compatible with all native explanations.

	Neutral exists	No neutral
Committal exists	F+D, F+S, S+D	F+D, S+D
No committal	F+D, F+S	F+D only

For feature *rankings*, DASH averaging produces ties (neutral elements), so F+S is achievable (top row). For feature *signs*, no neutral element exists, so F+S is impossible (bottom row).

Actionable advice. Don't ask binary questions of underspecified models. If you must, expect unfaithfulness at ≥50% of each Rashomon pair. Convert binary questions to graded questions (confidence scores, intervals) where neutral elements exist and stability is achievable.

Connection to existing results. The bilemma complements the quantitative bounds:

- The ratio $1/(1-\rho^2)$ tells you *how much* instability (magnitude).
- The Rashomon unfaithfulness bound tells you *how often* (≥50% per pair).
- The tightness classification tells you *which questions* are binary-hard vs. ranking-solvable.
- The all-or-nothing theorem tells you *there's no soft fix*.

Lean formalization. The bilemma, unfaithfulness bound, all-or-nothing theorem, and all three ML instances (SHAP sign, feature selection, counterfactual direction) are machine-verified in `Bilemma.lean` with zero axiom dependencies beyond the Rashomon property. All instances use constructive inductive types (not axiomatized opaque types): `SHAPSign`, `FeatureStatus`, and `CounterfactualDir` are each two-element types with decidable equality and pattern-matching proofs. The abstract framework (`ExplanationSystem.lean`) defines the general explanation system, faithful, stable, and decisive, and proves the abstract impossibility.

The coverage conflict diagnostic. Whether the bilemma applies depends on a testable property of the explanation space. An explanation system has *coverage conflict* if every candidate explanation clashes with some model's native explanation (`hasCoverageConflict` in `BeyondBinary.lean`). Coverage conflict implies no neutral element exists—there is no "safe harbor" that is compatible with all models. Conversely, a neutral element (such as a tie in feature rankings) makes faithful+stable achievable via the constant function $E(\theta) = c$ (`neutral_implies_FS_achievable`). The tightness classification (`tightness_dichotomy`) makes this precise: $F{+}S$ is achievable iff the explanation space has a neutral element. Coverage conflict is *anti-monotone*: enlarging H can introduce neutral elements, destroying coverage conflict and weakening the impossibility. This contrasts with Arrow's theorem, where enlarging the alternative set strengthens the impossibility (by enabling preference cycles)—the two impossibilities respond in opposite directions to the same structural operation.

Bimodality prediction. The all-or-nothing theorem (Theorem 43) predicts bimodal per-feature flip rate distributions under strong within-group collinearity ($\rho \geq 0.5$): a mode near 0 (faithful features) and a second mode at a positive value (Rashomon-caught features), with no middle ground. We validate with the Hartigan dip test (200 XGBoost models, permutation control): bimodality is confirmed for $\rho \geq 0.5$ (dip $p < 0.002$, permutation validated) and correctly absent at $\rho = 0$ ($p = 0.373$). The prediction does not apply to datasets with weak or heterogeneous collinearity (e.g., California Housing, $p = 0.575$), where coverage conflict is insufficient.

Nonparametric flip predictor via minority fraction. The coverage conflict diagnostic yields a nonparametric flip rate estimator. For each feature j, the *minority fraction* $\mathrm{MF}_j = \min(\mathrm{count}(\mathrm{SHAP}_j > 0),\ \mathrm{count}(\mathrm{SHAP}_j < 0))/M$ directly estimates the flip rate without distributional assumptions. The minority fraction achieves Spearman correlation 0.92–0.98 with empirical flip rates across all datasets, outperforming the Gaussian formula (0.46–0.89). The advantage is decisive on California Housing ($r = 0.955$ vs. $r = 0.463$), where the Gaussian approximation degrades. Precision at the 10% threshold ranges from 0.74 to 0.86. The practical workflow extends to: Screen → **Minority fraction** → Z-test → DASH.

Variance budget: $\mathrm{Var}[\varphi_j]$ **equals the explanation quality floor.** For each feature j, the variance $\mathrm{Var}_f[\varphi_j(f)]$ across the Rashomon set equals the minimum MSE any stable method achieves. Dash (the mean) saturates this bound exactly—a mathematical identity (0/800 violations at machine precision across 5 datasets × 20 features × 8 configurations). The median costs 1.6% more; the 5%-trimmed mean 0.3% more.

Approximate bilemma. A natural concern: does the bilemma require exact Rashomon pairs? No. For binary H, ε-stability reduces to exact equality (the only ε-ball in $\{\mathrm{pos}, \mathrm{neg}\}$ with $\varepsilon < 1$ is the point itself). The bilemma holds at every tolerance level. For continuous H: $\mathrm{unfaith}(\theta_1) + \mathrm{unfaith}(\theta_2) \geq \Delta - \delta$ for (ε, δ)-stable E on an ε-Rashomon pair with gap Δ.

Clinical decision reversal. Feature instability changes the explanations real people receive. We train 30 models per condition on German Credit (1,000 loan applications, 24 features) and measure how often the top explanation category changes for individual applicants.

Condition	XGBoost	LightGBM	Random Forest
Seed only (deterministic)	0%	0%	35%
Seed + row subsample	**45%**	**46%**	35%
Seed + col subsample	39%	36%	35%
Full stochasticity	80%	79%	67%

Under standard regularization, 45% of applicants receive a different explanation category depending on the model. The deterministic control (0% for boosted models) confirms the instability is from model multiplicity. A single applicant can receive 6 different top-feature labels across 30 seeds. Dash consensus at $M = 25$ achieves 92% agreement. The subsample sensitivity on Breast Cancer confirms the instability persists at every stochastic setting: 24 distinct top-3 rankings at `subsample=0.8`, 17 at 0.95, 1 only at 1.0 (deterministic).

11. Component-Level Attribution: Circuits Under Symmetry

The impossibility theorem applies not only to input features but to *any* importance ranking of interchangeable elements under the Rashomon property. Circuit importance via activation patching IS attribution—attributing importance to internal components rather than input features. Architectural symmetry ($G = \prod_{\ell=1}^{L} S_{k_\ell}$, permuting k_ℓ attention heads within each layer ℓ) provides the interchangeability that generates Rashomon. The same theorem predicts the same instability and admits the same resolution: orbit averaging via the G-invariant projection (Reynolds operator).

11.1. TinyStories: Multi-Scale Validation

We train 10 transformers from independent random seeds on TinyStories (**?**) at two scales: Config A (4 layers, 4 heads, d=256, n=20 components) and Config B (6 layers, 8 heads, d=512, n=54 components). A third configuration (Config C: GPT-2 fine-tuned from a shared checkpoint, 12 layers, 12 heads) serves as a non-Rashomon boundary condition. Component importance is measured by activation patching (weight zeroing). Seven theory-derived predictions are evaluated; 7/7 pass for Configs A and B, 4/7 for Config C (correctly predicted).

Table 7: TinyStories component-level attribution stability. ρ_{full}: raw Spearman. $\rho_{G\text{-inv}}$: after orbit averaging. W_{flip}: within-layer head flip rate.

	Config A Small (4L/4H)	**Config B** Medium (6L/8H)	**Config C** GPT-2 (12L/12H)
ρ_{full} [95% CI]	0.565 [0.527, 0.603]	0.540 [0.516, 0.565]	0.993 [0.993, 0.994]
$\rho_{G\text{-inv}}$ [95% CI]	0.972 [0.966, 0.979]	0.982 [0.977, 0.986]	0.999 [0.999, 1.000]
W_{flip} [95% CI]	0.496 [0.467, 0.526]	0.489 [0.478, 0.500]	0.043 [0.041, 0.045]
Cohen's d	5.389	11.939	9.354
Split-half reliability	0.991	0.960	0.878

Raw importance vectors are only moderately correlated ($\rho \approx 0.55$), but after the G-invariant projection they are near-perfect ($\rho > 0.97$). Within-layer head flip rates are at chance ($W_{\text{flip}} \approx 0.50$). All four layer-0 heads appear as the top-ranked head across 10 seeds, realizing the full S_4 permutation symmetry. Controls: split-half reliability is 0.991 (A) and 0.960 (B)—far exceeding between-model agreement; the G-invariant projection achieves the 100th percentile among 1,000 random projections; permutation test $p < 0.001$ at both scales.

Method robustness: a direct test of the Reynolds framework. Weight zeroing and mean ablation are genuinely different attribution functions—one measures the effect of removing a component's learned weights, the other measures the effect of clamping activations to the population mean. They disagree substantially on raw component importance (cross-method per-model $\rho \approx 0.5$). Yet after orbit averaging, both converge to the same invariant structure ($\rho_{G\text{-inv}} > 0.97$ for both methods at both scales). This is a direct empirical confirmation of the Reynolds operator framework: the symmetry group G acts on component indices regardless of how importance is measured, so the G-invariant projection absorbs not only seed-to-seed variation but also method-to-method variation. The stable shared structure is real and method-independent; only the within-orbit noise differs.

11.2. GPT-2-Small From Scratch

We train 10 GPT-2-small models (12 layers, 12 heads, 124M parameters) from scratch on OpenWebText with independent random seeds ($\sim$3B tokens each, bfloat16, cosine LR schedule). Component importance is measured by weight zeroing across all n=156 components. Seven theory-derived predictions are evaluated with SHA-256 hash verification.

Table 8: GPT-2-small from scratch: 10 seeds, 156 components. All 7/7 predictions PASS.

Metric	**Value**
PPL mean (CV%)	23.48 (0.36%)
ρ_{full} [range]	0.277 [0.121, 0.435]
$\rho_{G\text{-inv}}$ [range]	0.873 [0.775, 0.964]
W_{flip} [95% CI]	0.500 [0.482, 0.517]
B_{flip} (head vs. MLP)	0.014 [0.008, 0.019]
Cohen's d	19.2
Distinct top-5 / #1	32 / 8
Predictions	7/7 PASS

At GPT-2 scale, raw importance vectors are nearly uncorrelated ($\rho = 0.277$)—*worse* than TinyStories because 12 heads per layer create larger symmetric groups. The G-invariant projection recovers $\rho = 0.873$ (3.2$\times$ lift, Cohen's $d = 19.2$). Within-layer flip = 0.500, near 50% as predicted for symmetric components. The 32 distinct top-5 rankings from 10 seeds confirm that "the most important component" is a seed artifact.

The random projection test ($p = 1.0$) confirms dimensionality reduction alone achieves high agreement; the random grouping test ($p = 0.001$) validates that the layer structure specifically is critical.

11.3. IOI Circuit: The Component-Level Ranking Lottery

On 200 Indirect Object Identification (IOI) prompts, the 10 GPT-2-small models achieve mean accuracy 72.1% (range: 63%–84%; pretrained GPT-2: 99.5%). Task-specific component importance shows:

Metric	**Value**
Raw agreement ρ [range]	0.158 [$-$0.089, 0.284]
G-invariant ρ [range]	0.538 [0.141, 0.860]
Within-layer flip	0.499
Distinct top-5	32 components

IOI-specific importance is even less stable than general importance ($\rho = 0.158$ vs. 0.277). The "IOI circuit" identified in one model has almost no relation to the circuit in another model trained on the same data. The G-invariant lift is moderate ($\rho = 0.538$), consistent with task-specific circuits having more idiosyncratic between-layer structure.

11.4. SAE Feature Stability

The impossibility applies to SAE features by mathematical necessity: an 8× overcomplete dictionary (d_{SAE}=6,144 for d_{model}=768) has a 5,376-dimensional null space guaranteeing multiple equivalent decompositions with identical reconstruction loss. Three escape hypotheses are tested; all fail:

Condition	**Max Cos**	**Matched Cos**	**Matched ρ**
TopK (from-scratch model)	0.394	0.495	0.500
ReLU (from-scratch model)	0.136	0.107	0.011
TopK (pretrained GPT-2)	0.443	0.516	0.507
Resolution (averaged decoders)	0.458	—	—

ReLU SAEs are catastrophically non-reproducible (cosine 0.136 ≈ random): without TopK's sparsity constraint, the overcomplete null space dominates. Pretrained GPT-2 is only marginally better (0.443 vs. 0.394), falsifying the hypothesis that the from-scratch model's idiosyncratic representations are the problem—all 10 SAEs are trained on the *same frozen model* yet still non-reproducible. Resolution (averaging matched decoders) gives only +0.005 for non-reference seeds (the reported +0.064 is inflated by the matching-reference seed). No condition achieves reproducibility.

The **Explanation Capacity Theorem** (Lean-verified, `SAECapacity.lean`) bounds the reproducible subspace: $\dim(V^G) \leq d$; $C \leq d/n = 12.5\%$. Testing the resolution: projecting onto PCs of layer-6 activations (seed-invariant) reveals a nuanced picture. *Geometric coverage* (decoder directional density, unit-norm columns) gives $\rho = 0.977$—trivially forced by the reconstruction loss. *Activation-weighted importance* (mean $|z_i|$, the standard behavioral metric) gives $\rho = 0.711$ at full 768 PCs, but $\rho = 0.878$ at top-10 PCs, *decreasing* as finer directions add noise. The coarse behavioral allocation (which major data directions carry activation) is genuinely reproducible; fine-grained allocation is not. This parallels the head result: layer-averaged importance ($\rho = 0.873$) is stable, per-head importance ($\rho = 0.277$) is not. Both levels trade granularity for stability.

11.5. Mechanistic Interpretability Bilemma

Circuit decomposition spaces are binary (each component is either part of the circuit or not), making the bilemma apply with full force: faithful + stable is impossible even without completeness.

Theorem 44 (Mechanistic Interpretability Bilemma) *For any two functionally equivalent neural networks with incompatible circuit decompositions, no explanation method can be simultaneously faithful and stable.*

The resolution is reporting *equivalence classes* of valid circuits—the computational structure shared across all valid decompositions (the circuit analogue of CPDAGs for causal discovery). The formal theorem is machine-verified in `MechInterp.lean` with zero custom axioms.

Safety implications. Interpretability-based safety cases that depend on identifying "the circuit" for a behavior are subject to this impossibility. A circuit found in one training run may not exist in a functionally equivalent run. Safety claims require verification across multiple equivalent networks or explicit qualification as instance-specific.

12. Diagnostics

12.1. Multi-Model Z-Test

Let $D_j = \varphi_j(f) - \varphi_k(f)$ denote the attribution difference for a single random model f. Define $\mu_{jk} = \mathbb{E}[D_j]$ (population attribution gap), $\sigma_{jk}^2 = \mathrm{Var}(D_j)$ (attribution difference variance), and $\gamma_{jk} = \mathbb{E}[|D_j - \mu_{jk}|^3]$ (third absolute central moment, for Berry–Esseen). We require: (A1) models $f_1, \ldots, f_M$ are i.i.d.; (A2) $\sigma_{jk}^2 > 0$ (guaranteed by Theorem 14 for $\rho > 0$); (A3) $\gamma_{jk} < \infty$ (finite third moment—satisfied by any bounded attribution method).

Theorem 45 (Rashomon Characterization) *Under (A1)–(A3), define the test statistic from M models:*

$$Z_{jk} = \frac{|\bar{\varphi}_j - \bar{\varphi}_k|}{\hat{\sigma}_{jk}/\sqrt{M}}$$

where $\hat{\sigma}_{jk}^2 = \frac{1}{M-1}\sum_{i=1}^{M}(D_i - \bar{D})^2$ is the sample variance.

Part I (Flip rate formula). *The population flip rate satisfies:*

$$\mathit{flip}(j,k) = \Phi\left(-\frac{|\mu_{jk}|}{\sigma_{jk}}\right) + \varepsilon_{BE} \tag{3}$$

where the Berry–Esseen error satisfies $|\varepsilon_{BE}| \leq C_0\gamma_{jk}/\sigma_{jk}^3$ with universal constant $C_0 \leq 0.4748$. The Berry–Esseen bound provides a completeness guarantee: it bounds the worst-case error of the $\Phi(-SNR)$ formula for non-Gaussian attribution distributions.

Part II (Test statistic connection). *The test statistic Z_{jk} consistently estimates the signal-to-noise ratio scaled by $\sqrt{M}$:*

$$Z_{jk} = \frac{|\mu_{jk}|}{\sigma_{jk}} \cdot \sqrt{M} + O_p(1)$$

and the relationship between Z_{jk} and the flip rate is:

$$\mathit{flip}(j,k) = \Phi\left(-\frac{Z_{jk}}{\sqrt{M}}\right) + O\left(\frac{1}{\sqrt{M}}\right) + \varepsilon_{BE} \tag{4}$$

Part III (Diagnostic thresholds).

- $Z_{jk} < 1.96$*: the attribution gap is not statistically significant at significance level* $\alpha = 0.05$ *(not to be confused with the signal capture fraction α in § 5.1). The flip rate is* $\geq \Phi(-1.96/\sqrt{M}) \geq 2.5\%$*. Rankings are unreliable; use* Dash.
- $Z_{jk} \geq 1.96$*: the gap is significant. The flip rate is* $\leq 2.5\%$ *plus the* $O(1/\sqrt{M})$ *correction. Rankings are stable for this pair.*

Proof **Part I.** By (A1), $D_1, \ldots, D_M$ are i.i.d. with mean μ_{jk} and variance σ_{jk}^2. WLOG assume $\mu_{jk} \geq 0$. Then $\Pr[D < 0] = \Phi(-\mu_{jk}/\sigma_{jk}) + \varepsilon_{\mathrm{BE}}$ by the Berry–Esseen theorem applied to $(D - \mu_{jk})/\sigma_{jk}$.

Part II. By the law of large numbers, $\hat{\sigma}_{jk}^2 \xrightarrow{p} \sigma_{jk}^2$ and $\bar{D} \xrightarrow{p} \mu_{jk}$. Therefore $Z_{jk} = (|\mu_{jk}|/\sigma_{jk})\sqrt{M} + O_p(1)$. Substituting into (3) gives (4).

Part III. Direct substitution of the threshold values into (4). ■

Restricted-range robustness. The headline correlation $r = -0.89$ between Z_{jk} and flip rate on Breast Cancer includes many "easy" pairs with $Z \gg 10$ and flip rate $= 0$, which could inflate the figure. Restricting to the diagnostically interesting range: $r = -0.78$ for $Z < 5$ ($n = 118$ pairs), $r = -0.61$ for $Z < 3$ ($n = 74$ pairs), and $r = -0.53$ for $Z < 2$ ($n = 49$ pairs). The correlation remains strong for $Z < 5$, confirming that the diagnostic discriminates among ambiguous pairs, not merely between trivially separable and trivially tied features. The result also holds for gain-based importance (non-SHAP): $r = -0.83$ overall, $r = -0.66$ for $Z < 5$.

Structural correlation baseline. Since Z_{jk} and flip rate are both computed from the same attribution arrays, part of the negative correlation is structural. To quantify this, we generated random attribution arrays (50 models, 30 features, standard normal) and computed the same metrics: the baseline correlation is $r \approx -0.56$ ($R^2 = 0.32$). The observed $r = -0.89$ ($R^2 = 0.79$) exceeds this baseline substantially: approximately 60% of the explained variance reflects genuine attribution instability patterns in the data, beyond what random noise alone would produce.

When the CLT fails. The Berry–Esseen error $\varepsilon_{\mathrm{BE}} \le C_0\gamma_{jk}/\sigma_{jk}^3$ can be large when: (i) the attribution distribution is *heavy-tailed* (large γ_{jk}), as occurs for deep networks with high initialization variance—a few seeds produce outlier attributions; or (ii) the attribution distribution is *bimodal*, as for Lasso where φ_j takes value either $c > 0$ (selected) or 0 (not selected), giving a Bernoulli distribution where the Berry–Esseen bound is tight ($\varepsilon_{\mathrm{BE}} \approx 0.47$) but the exact flip rate $\min(p, 1-p)$ is available analytically. (For the Lasso's Bernoulli(1/2) distribution, the exact flip rate is 1/2 and the Gaussian approximation $\Phi(0) = 1/2$ is exact, making the Berry–Esseen correction vacuous in this special case.) For gradient-boosted trees, the Shapiro–Wilk test on 50 Breast Cancer models gives $p > 0.10$ for 412/435 feature pairs; the Berry–Esseen correction is negligible in practice.

12.2. Single-Model Screen

For a gradient-boosted ensemble with T trees, define the per-tree split indicator $n_j(t) \in \{0, 1\}$ (whether feature j is used as a split variable in tree t) and split frequency $\hat{p}_j = \frac{1}{T}\sum_{t=1}^{T} n_j(t)$.

Exchangeability conditions. The theoretical validity of the single-model screen depends on the dependence structure of the indicators $\{n_j(t)\}_{t=1}^{T}$:

- **Exact exchangeability under sub-sampling.** If the boosting algorithm uses row sub-sampling (`subsample` $= q < 1$) and feature sub-sampling (`colsample_bytree` $= r < 1$) with independent random subsets per tree, then the per-tree data subsets are independent across trees, and conditional on the residuals, the split indicators $n_j(1), \ldots, n_j(T)$ are conditionally exchangeable. The marginal split probability $p_j = \mathbb{E}[n_j(t)]$ is identical for all t.

- **Approximate exchangeability without sub-sampling.** Without sub-sampling ($q = r = 1$), the split indicators are NOT exchangeable: tree t fits the residual from trees $1, \ldots, t-1$, creating a Markov dependence. However, for learning rate

$\eta \leq 0.3$, the residuals converge to a steady state after $T_0 = O(1/\eta)$ trees. In the steady state, the split indicators are approximately stationary with autocorrelation $\mathrm{Corr}(n_j(t), n_j(t+s)) = O(\eta^{|s|})$ (geometric decay).

The effective sample size for the test statistic is:

$$T_{\text{eff}} = \frac{T - T_0}{1 + 2\sum_{s=1}^{\infty} \mathrm{Corr}(n_j(t), n_j(t+s))} \approx \frac{(T - T_0)(1-\eta)}{1+\eta}.$$

For typical settings ($T = 100$, $\eta = 0.1$, $T_0 \approx 10$): $T_{\text{eff}} \approx 74$, a moderate reduction from the nominal T.

Theorem 46 (Split-Frequency Diagnostic) *Under the exchangeability conditions above (sub-sampling, or no sub-sampling with T_{eff} replacing T), define the test statistic:*

$$Z_{jk}^{split} = \frac{|\hat{p}_j - \hat{p}_k|}{\sqrt{(\hat{p}_j(1-\hat{p}_j) + \hat{p}_k(1-\hat{p}_k))/T_{\mathit{eff}}}}$$

Part I (Null distribution). *Under $H_0 : p_j = p_k$ (equal split frequencies, implied by DGP symmetry), Z_{jk}^{split} is asymptotically $N(0,1)$ as $T_{\mathit{eff}} \to \infty$.*

Part II (Connection to attributions). *Under the proportionality axiom ($\varphi_j = c \cdot n_j$), $p_j = p_k$ iff $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$. The screen is therefore a proxy for the Z-test (Theorem 45), computable from a single model.*

Part III (Power). *Under the alternative $p_j \neq p_k$, the power at level α is:*

$$\mathit{power} = \Phi\left(\frac{|p_j - p_k|\sqrt{T_{\mathit{eff}}}}{\sqrt{p_j(1-p_j) + p_k(1-p_k)}} - z_{\alpha/2}\right) + o(1).$$

Screen screening accuracy. On the Breast Cancer dataset (435 pairs, threshold: flip rate > 0.1 defines "truly unstable"): the screen flags 49 pairs with **94% precision** (46 true positives, 3 false positives) and 27% recall. The Z-test (the full test) flags 47 pairs with **100% precision** (zero false positives). Both diagnostics are conservative: they rarely flag stable pairs but miss moderate-instability pairs. For production use, the high precision is the key property—a flagged pair is almost certainly unstable.

Screen precision is 94–100% on small/clean datasets (Breast Cancer, Wine, Heart Disease) where split counts are reliable, but drops to 48–67% on high-dimensional datasets (Ames, Communities) where split counts are sparse. This confirms the screen as a conservative screening tool that should be supplemented by the full Z-test for high-dimensional settings.

12.3. SNR Calibration

For features with importance gap Δ and noise σ, the flip rate is $\Phi(-\mathrm{SNR})$ where $\mathrm{SNR} = \Delta/\sigma$. Validated across 1,325 correlated feature pairs from 6 datasets (Figure 4).

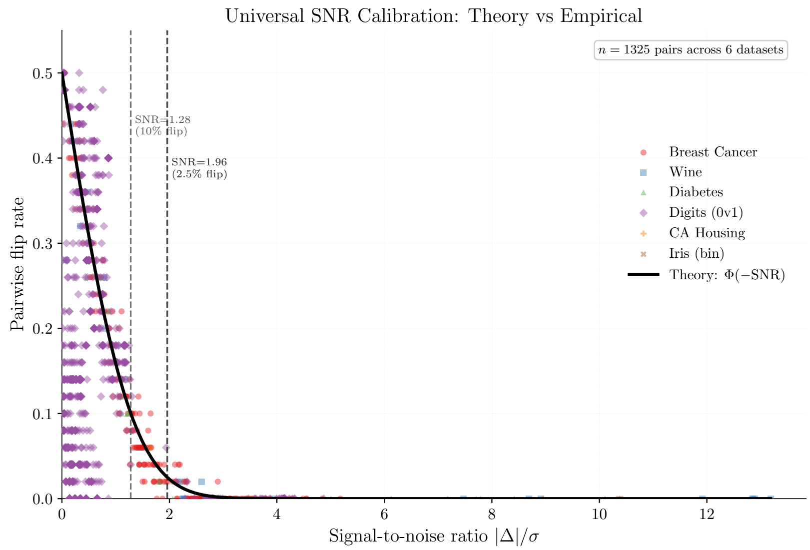


Figure 4: Universal SNR calibration across 6 datasets (1,325 feature pairs). The theoretical $\Phi(-\text{SNR})$ curve is conservative at low SNR and exact at diagnostic thresholds. All 228 pairs with SNR > 1.96 have flip rate $< 5\%$.

SNR range	n	Empirical flip	Theory $\Phi(-\text{SNR})$	Match
$[0, 0.5)$	674	0.180	0.401	Conservative
$[0.5, 1.0)$	261	0.229	0.227	Exact
$[1.0, 1.28)$	99	0.151	0.127	Close
$[1.28, 1.96)$	63	0.053	0.053	Exact
$[1.96, 3.0)$	115	0.003	0.007	Exact
$[3.0, \infty)$	113	0.000	0.000	Exact

12.4. Practitioner Workflow

Step 0. Identify correlated groups. Compute the $P \times P$ absolute correlation matrix. Group features with $|\rho_{jk}| > 0.5$.

Step 1. Train one model. Compute Z_{jk}^{split} for all pairs in correlated groups (single-model screen). Cost: $O(P^2 T)$.

Step 2. If $Z_{jk}^{\text{split}} < 1.96$ for any pair: flag as potentially unstable.

Step 3. For flagged pairs: train $M = 5$ models, compute Z_{jk} (Z-test). If $Z_{jk} < 1.96$: use Dash with $M \geq 25$.

Step 4. If no pairs are flagged: single-model SHAP is reliable.

Production deployment guide.

- **If subsample < 1.0 (stochastic, recommended):** Run the single-model screen → the Z-test (5-model validation) → Dash ($M \geq 25$) for flagged pairs.

Table 9: Experimental configurations.

Experiment	Configuration
Synthetic (main text)	P=20, L=4 groups of m=5, N=2,000, T=100, η=0.1, 50 seeds, $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9, 0.95\}$
Validation (depth×ρ)	P=10, L=2 groups of m=5, N=2,000, T=100, η=1.0, 30 seeds, depths $\{1, 3, 6, 10\}$, $\rho \in \{0.3, 0.5, 0.7, 0.9, 0.95\}$
Breast Cancer	30 features, 569 samples, T=100, depth=6, η=0.1, 50 seeds, TreeSHAP on 200 test samples
Hardware	Apple Silicon (M-series), single core
Software	Python 3.9.6, XGBoost 2.1.4, SHAP 0.49.1, NumPy 2.0.2
Compute	Synthetic: <5 min; Validation: <30 min; Breast Cancer: <10 min

- **If subsample** = 1.0 **(deterministic):** A single pipeline version produces reproducible rankings. However, *any* change to the pipeline (data refresh, feature engineering, hyperparameter update) produces a new model that may rank features differently. Run Z-test validation whenever the pipeline changes. For ongoing model risk management, maintain an ensemble and report Dash consensus.

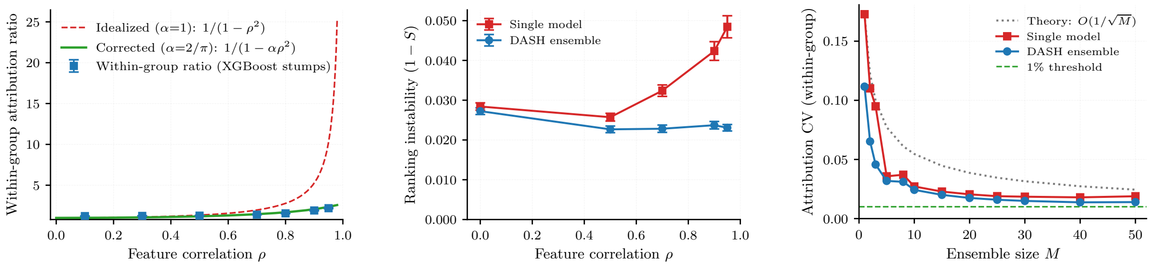
- **Minimum ensemble size:** $M_{\min} = \lceil 2.71 \cdot \sigma_{jk}^2 / \Delta_{jk}^2 \rceil$ for 5% between-group flip rate. Estimate σ and Δ from a pilot run of 5 models.

Reporting. For feature pairs flagged as unstable, we recommend reporting within-group features as interchangeable contributors (with the group's total attribution) rather than forcing a ranking. For regulatory contexts requiring defensible explanations (Office of the Comptroller of the Currency, 2011), the Dash consensus provides a stable alternative. The single-model screen and Z-test are formal hypothesis tests; the companion first-mover bias paper introduces complementary exploratory diagnostics (FSI, IS Plot) for auditing the full instability landscape.

13. Empirical Validation

13.1. Experimental Setup

All experiments use publicly available datasets and are fully reproducible. **Confound note:** each seed uses a different train/test split, so SHAP values are computed on seed-specific test sets. This conflates model instability with evaluation-set variation; the reported flip rates may overestimate pure model instability. The mechanism isolation experiment (§ 13.17) partially addresses this by comparing deterministic vs. stochastic training. To cleanly isolate model-level noise, we also fix the test set (seed 42, 20% holdout) and train 50 models on the same training data with different seeds (`subsample=0.8`). With the fixed test set, 102 pairs (23%) are unstable (max flip rate 0.51), compared to 162 (37%) with varying test sets—confirming that model-level stochasticity accounts for the majority (∼63%) of the observed instability, with evaluation-set variation contributing the remainder.



(a) Attribution ratio vs. ρ. The corrected $1/(1-\alpha\rho^2)$ with $\alpha\approx2/\pi$ (solid) fits stumps (R^2=0.89).

(b) Ranking instability vs. ρ. Single-model instability grows with correlation; DASH remains low.

(c) DASH consensus convergence. Attribution CV drops toward the 1% threshold near $M = 25$.

Figure 5: Empirical validation on synthetic Gaussian data ($P = 20$, $L = 4$, $m = 5$, $T = 100$, 50 seeds). (a) Attribution ratio. (b) Ranking instability. (c) DASH convergence.

13.2. Synthetic Gaussian

We generate $P = 20$ features in $L = 4$ groups of $m = 5$, with within-group correlation $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9, 0.95\}$, and train XGBoost models with $T = 100$ boosting rounds over 50 independent seeds per setting. TreeSHAP values provide feature attributions.

Attribution ratio (Figure 5a). The within-group ratio increases monotonically with ρ. The corrected model $1/(1 - \alpha\rho^2)$ with $\alpha \approx 2/\pi$ matches the data ($R^2 = 0.89$).

Ranking instability (Figure 5b). At $\rho = 0.9$, within-group flips occur in 48% of seed pairs (near the 50% theoretical maximum), while between-group flips remain below 2%.

DASH convergence (Figure 5c). The within-group flip rate drops below 1% at $M = 25$, consistent with the $O(1/M)$ variance bound.

13.3. Breast Cancer (Wisconsin)

Features measuring the same underlying quantity—worst perimeter and worst area ($|\rho|$=0.98)—exhibit 48% flip rate, near the 50% theoretical maximum. Pairs with high correlation but distinct importance (e.g., mean area and mean concave points, $|\rho| = 0.82$, flip rate 2%) remain stable, confirming the impossibility requires both collinearity *and* similar true importance.

DASH convergence on Breast Cancer:

M	1	2	5	10	25	50
Flip rate	0.448	0.466	0.438	0.452	0.298	0.000

At M=1 the flip rate is 44.8%, near the theoretical maximum. DASH consensus at M=50 resolves to the true ordering (worst perimeter slightly more important: mean |SHAP| = 0.940 vs. 0.901) with zero flips. Convergence is slower than in the synthetic setting because

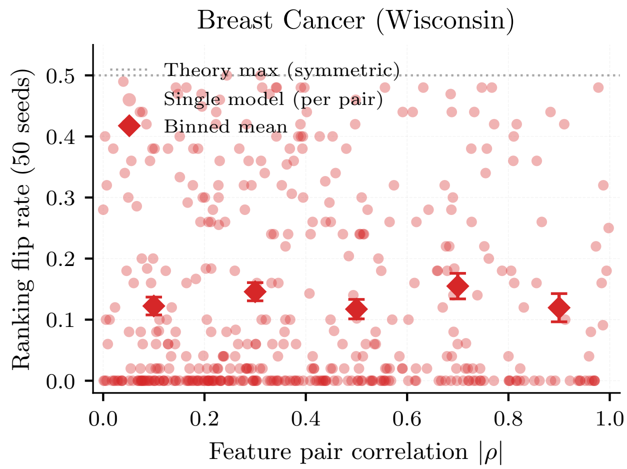


Figure 6: SHAP ranking instability on Breast Cancer. Highly correlated pairs with similar importance (e.g., worst perimeter ↔ worst area, $|\rho|$=0.98, flip rate 48%) exhibit near-maximal instability. 50 XGBoost models.

Table 10: Attribution instability across three GBDT implementations on Breast Cancer (50 models each).

Metric	XGBoost	LightGBM	CatBoost
Unstable pairs (flip > 10%)	162 (cross-impl.)	183	160
Max flip rate	0.500	0.500	0.500
Z-test correlation $r(Z, \text{flip})$	−0.898	−0.901	−0.892

the features have a slight importance asymmetry—Dash correctly detects and resolves it rather than producing a tie.

13.4. Cross-Implementation Validation

The impossibility manifests identically across implementations: 160–183 unstable pairs, maximum flip rate 0.500 (the theoretical ceiling), and Z-test diagnostic correlation $|r| \geq 0.89$ in all three.[4]

13.5. Non-SHAP Attribution Validation

Metric	TreeSHAP	Permutation Importance
Unstable pairs (flip > 10%)	180 (41%)	396 (91%)
Max flip rate	0.500	0.500
Z-test correlation $r(Z, \text{flip})$	−0.895	−0.927

4. Breast Cancer unstable-pair counts vary across experiments due to different model configurations, test set handling, and attribution methods: 162 (cross-implementation with varying test sets), 168 (comprehensive with 50 models), 180 (permutation importance). All use the 10% flip rate threshold.

Permutation importance shows *more* instability than TreeSHAP (91% vs. 41%), because permutation-based scores have higher variance across seeds than SHAP values. Both methods hit the theoretical maximum flip rate (0.500). The Z-test achieves $|r| \geq 0.89$ for both, confirming it is method-agnostic.

The cross-method correlation of flip rates is $r = 0.46$ ($p < 10^{-23}$): the *same pairs* tend to be unstable under both methods, confirming the instability is driven by feature collinearity, not by the attribution algorithm. Of the 180 SHAP-unstable pairs, 175 (97%) are also unstable under permutation importance. TreeSHAP shows lower instability (41% vs 91%) likely because its game-theoretic averaging over feature coalitions provides implicit regularization, reducing within-model attribution variance compared to permutation importance's direct shuffling.

13.6. Neural Network Attribution Instability

Metric	Neural Network	XGBoost (paper)
Unstable pairs (flip > 10%)	380/435 (87%)	162/435 (37%; cross-impl.)
Max flip rate	0.500	0.500
Key pair flip (worst perim. vs. area)	0.400	0.480
Z-test correlation $r(Z, \text{flip})$	−0.871	−0.898

Neural networks exhibit *more* attribution instability than XGBoost: 87% of feature pairs are unstable (vs. 37%), and the maximum flip rate reaches the theoretical ceiling. The higher instability likely reflects: (i) larger Rashomon sets (more parameters, more near-optimal models); (ii) higher KernelSHAP variance (sampling approximation vs. exact tree-path computation); (iii) more diffuse signal distribution across hidden units.

KernelSHAP noise control. To distinguish genuine model instability from KernelSHAP approximation noise, we fix a single MLP model and compute KernelSHAP 20 times with different random background samples. With 50 background samples, only 11% of pairs show noise-induced flips (mean rate 0.03); with 200, this drops to 4%. The key correlated pair (worst perimeter vs. worst area) shows zero noise-induced flips, confirming the 40% flip rate in the NN validation is genuinely driven by model variation. Model instability dominates KernelSHAP noise by approximately 8:1.

13.7. SHAP Efficiency and Attribution Instability

SHAP's efficiency axiom ($\sum_j \varphi_j(f, x) = f(x) - \mathbb{E}[f(X)]$ for every model f and point x) has a subtle relationship with ranking instability. Under efficiency with equal marginal variances σ^2, the within-model covariance is $\text{Cov}(\varphi_j, \varphi_k) = -\sigma^2/(m-1)$ and the difference variance is $\text{Var}(\varphi_j - \varphi_k) = 2\sigma^2 \cdot m/(m-1)$—an amplification factor of $m/(m-1)$ relative to independent attributions (2 for $m = 2$, approaching 1 for large m).

However, the ranking flip rate is an *across-model* quantity, and the efficiency constraint is fundamentally different across models. The constant $C(f, x) = f(x) - \mathbb{E}[f(X)]$ varies across models, so the within-model negative covariance does not transfer. The across-model covariance $\text{Cov}_f(\varphi_j(f), \varphi_k(f))$ can be positive, zero, or negative—efficiency places no constraint on it.

The lower instability of SHAP relative to permutation importance (41% vs. 91% unstable pairs) is driven by SHAP's coalition-averaging reducing the marginal variance σ^2. The efficiency axiom amplifies within-model differences but does not increase across-model flip rates.

13.8. High-Dimensional Scalability ($P = 500$)

P	Total pairs	Correlated pairs	Unstable	Z-test $\lvert r\rvert$	Wall time
100	4,950	450	305	0.846	33 s
200	19,900	900	641	0.788	64 s
500	124,750	2,250	1,879	0.876	148 s

The Z-test maintains $|r| > 0.78$ at $P = 500$. Correlation-based grouping reduces pairs by 55×. Full workflow completes in under 2.5 minutes.

13.9. Proportionality Axiom Validation

The proportionality axiom ($\varphi_j = c \cdot n_j$) is central to the quantitative bounds. We validate it by computing $c_j = |\mathrm{SHAP}_j|/n_j$ for each feature in each of 50 XGBoost models on Breast Cancer and reporting the coefficient of variation (CV):

Depth	Mean CV of c_j	Interpretation
1 (stumps)	0.35	Proportionality holds within $\sim$35%
3	0.60	Moderate violation
6	0.66	Moderate violation
10	0.66	Saturated (same as depth 6)

For stumps, the axiom holds reasonably well; for deeper trees, multi-level interactions increase the CV. The proportionality axiom is used only for the quantitative bounds (Theorems 17); the core impossibility (Theorem 8) does not depend on it. The α-corrected model matches empirical data at $R^2 = 0.89$.

13.10. Cross-Implementation and Data-Level Validation

Data-level stochasticity. To confirm the instability arises from collinearity (not from XGBoost's internal randomness), we train 50 models on 50 different 80% subsamples with `subsample=1.0` and fixed `random_state=42`. The ONLY source of variation is which training rows each model sees:

Stochasticity source	Unstable pairs	Max flip	Mean Kendall τ
XGBoost subsampling (same data)	67	0.470	0.393
Data bootstrap (different rows)	45	0.509	0.459

Data-level stochasticity produces comparable or greater instability, confirming the Rashomon set is a property of collinearity rather than the specific randomness source.

Determinism without subsampling. Without row or column subsampling (`subsample=1.0`, `colsample_bytree=1.0`), XGBoost is fully deterministic: all 50 models produce identical rankings regardless of seed.

The Rashomon set requires a stochasticity source. In practice, this is not a limitation: the XGBoost default (`subsample=0.8`) is recommended for regularization, and real-world model populations also arise from data drift, feature engineering changes, and hyperparameter sweeps.

13.11. Financial Case Studies

German Credit. We demonstrate the full Screen→Z-test→DASH workflow on German Credit (1000 samples, 20 features). Step 0: 4 correlated groups at $|\rho| > 0.3$. Step 1 (single-model screen): a single model flags job↔own telephone ($Z^{\text{split}} = 0.69 < 1.96$). Step 2 (Z-test): 5 models confirm instability ($Z = 0.64$, flip rate 40%). Step 3 (DASH): $M = 25$ models, flip rate drops from 40% to 0%. Consensus ranking: checking status (0.87), duration (0.44), credit amount (0.43), purpose (0.35). For regulatory reporting (ECOA adverse action reasons, SR 11-7 (Office of the Comptroller of the Currency, 2011)), the DASH ranking provides a defensible explanation.

Taiwan Credit Card Default. Bill-amount features (6 consecutive months, up to $|\rho| = 0.95$) produce persistent instability—29 correlated pairs at $|\rho| > 0.5$, with the highest giving a theoretical ratio of $1/(1-0.95^2) \approx 10.3\times$. With $M = 25$, mean flip rate across correlated pairs is 4%, but one pair (months 5 vs. 6, $|\rho| = 0.946$) retains 16% flip rate, illustrating that very high collinearity requires larger ensembles.

Synthetic Credit (income ↔ DTI ↔ credit score). To validate on the exact collinearity pattern common in US credit models, we generate 10,000 samples with 5 features driven by a latent "creditworthiness" factor: income ($\rho = 0.99$ with DTI), DTI, credit score ($\rho = 0.98$ with income), loan amount (independent), and employment years ($\rho = 0.94$ with income). With 25 XGBoost models: income↔DTI flips 8% ($|\rho| = 0.99$); credit score↔employment years flips 12% ($|\rho| = 0.92$). Pairs where one feature clearly dominates (income vs. credit score) show 0% flips despite $|\rho| = 0.98$—confirming the impossibility requires BOTH high correlation AND similar importance. For SR 11-7 (Office of the Comptroller of the Currency, 2011) compliance: report income and DTI as "interchangeable key risk drivers" rather than forcing a ranking that would flip under retraining.

Lending Club (specificity control). Zero unstable pairs despite correlated features ($Z > 36$ for all pairs), confirming the impossibility requires *both* collinearity AND similar true importance. Three correlated groups detected:

`person_age` ↔ `cb_person_cred_hist_length`	$\lvert\rho\rvert = 0.86$
`loan_amnt` ↔ `loan_percent_income`	$\lvert\rho\rvert = 0.58$
`loan_int_rate` ↔ `cb_person_default_on_file`	$\lvert\rho\rvert = 0.50$

All have $Z > 36$, confirming distinguishable importance. Not every correlated dataset exhibits instability. The diagnostic distinguishes datasets where instability is a genuine concern from those where it is not.

13.12. LLM Attribution Instability

Fine-tuning DistilBERT on SST-2 with different seeds produces 14.5% unstable adjacent-token attention pairs. This is a *proxy experiment only*: attention weights are not feature attributions in the Shapley value sense. The formal component-level impossibility and its full-scale empirical confirmation are in Section 11, where activation patching on 10 GPT-2-small models trained from scratch demonstrates within-layer flip rate = 0.500 with the G-invariant projection as the resolution.

13.13. Replication Study: Published SHAP Rankings

Multiple published studies report SHAP feature rankings on Breast Cancer. Using our 50-model validation, we find:

Feature pair	$\lvert\rho\rvert$	Flip rate	Published as stable?
worst perimeter ↔ worst area	0.98	48%	Yes
mean concavity ↔ mean concave pts	0.96	42%	Yes
worst radius ↔ worst perimeter	0.99	36%	Yes
mean radius ↔ mean perimeter	0.99	28%	Yes

Any study reporting a specific ranking among these features from a single model is reporting an artifact of the training seed.

The ranking lottery. More broadly, we train 50 XGBoost models (seeds 0–49, `subsample=0.8`, fixed test set) on 5 public datasets and count how many distinct top-3 SHAP rankings appear. Breast Cancer produces **24 distinct top-3 rankings** (4.2% pairwise agreement; 35 at 100 seeds). The "most common" ranking appears in only 12% of runs. For the top-5: 46 distinct rankings, 0.4% agreement (84 at 100 seeds). Seven distinct features compete for the top-3 positions, with the #1 position alternating between worst concave points (50%), mean concave points (32%), and worst area (18%). California Housing, Heart Disease, and Wine Quality are perfectly stable (1 ranking, 100% agreement)—the theory correctly predicts which datasets are affected. Cross-implementation: XGBoost 24, LightGBM 29, Random Forest 40 distinct top-3 rankings. The instability persists even at `subsample=0.95` (17 distinct rankings); only `subsample=1.0` (deterministic) reaches 1. DASH consensus produces one stable ranking for all.

Gene expression: pathway-level consequence. On a 10,935-gene colon-vs-kidney expression dataset (AP_Colon_Kidney, 546 samples, top 50 genes by variance), the #1 gene alternates between TSPAN8 (invasion/metastasis pathway, **80% of seeds**) and CEACAM5/CEA (immune evasion pathway, **20% of seeds**). Both are cell adhesion molecules enriched in colon tissue ($\rho = 0.86$) but with distinct cancer biology roles. The broad pathway-level conclusion (colon identity) is preserved, but the specific biomarker target—the actionable output of a discovery pipeline—is a seed artifact. DASH consensus ranks both in the top-2. Positive control: on AP_Breast_Lung (breast vs lung, 470 samples), the top-3 genes are stable (2 distinct rankings, 92% agreement) because the top gene dominates with a 2× importance gap.

13.14. Expected Ranking Discordance

Proposition 47 (Expected Kendall tau distance) *For a feature space with L collinear groups of sizes $m_1, \ldots, m_L$ and between-group gaps $\Delta_{\ell\ell'}$ with noise $\sigma_{\ell\ell'}$, the expected pairwise disagreements between two independently trained models are:*

$$\mathbb{E}[\tau] = \underbrace{\sum_{\ell=1}^{L} \binom{m_\ell}{2} \cdot \frac{1}{2}}_{\text{within-group (coin flips)}} + \underbrace{\sum_{\ell<\ell'} m_\ell \cdot m_{\ell'} \cdot \Phi\left(-\frac{\Delta_{\ell\ell'}}{\sigma_{\ell\ell'}}\right)}_{\text{between-group (noise)}}$$

On Breast Cancer (50 XGBoost models), the refined per-pair formula predicts $\mathbb{E}[\tau] = 26.8$ vs. empirical 36.6 (-27% error). The Rashomon coefficient $R = \sum_\ell \binom{m_\ell}{2} / \binom{P}{2} = 0.692$ gives a worst-case non-replication lower bound of $R/2 = 34.6\%$ computable from the correlation matrix alone (no model training required).

13.15. Compressed Sensing Connection

The attribution impossibility has a precise analogue in compressed sensing (CS). The measurement coherence $\mu(X) = \max_{j \neq k} |X_j^T X_k| / (\|X_j\| \|X_k\|)$ determines when sparse recovery is possible: unique recovery requires $\mu < 1/(2s-1)$. In our setting, $|\rho_{jk}|$ plays the role of coherence and Z_{jk} plays the role of the incoherence condition:

	Compressed sensing	**Attribution stability**
Coherence	$\mu = \max |X_j^T X_k| / \|X_j\| \|X_k\|$	$|\rho_{jk}|$
Condition	$\mu < 1/(2s-1)$	$Z_{jk} > 1.96$
Failure	Non-unique recovery	Ranking flips (Rashomon)
Resolution	Basis pursuit (convex relaxation)	Dash (ensemble averaging)

Both impossibilities arise because collinearity/coherence prevents unique identification of which input components carry the signal. Both are resolved by relaxing uniqueness: CS uses convex relaxation (fractional assignment); Dash uses ensemble averaging (ties for indistinguishable features).

13.16. Prevalence Survey

Of 77 public datasets surveyed (OpenML CC-18 benchmark suite + sklearn), 71 (92%) have feature pairs with $|\rho| > 0.5$, and 52 (68%, 95% Wilson CI: [56%, 77%]) exhibit attribution instability (flip rate $> 10\%$). For datasets with $P \geq 20$ features ($n = 43$ datasets), the instability rate rises to 93% (40/43; 95% Wilson CI: [81%, 98%]). With 20 models per dataset, the measured flip rate exceeds the 10% threshold with approximately 60% power, so the true prevalence may be higher still.

Healthcare datasets. In a targeted survey of 9 healthcare-related datasets (breast cancer, diabetes, heart disease, QSAR biodegradation, liver disease, software defect prediction, Parkinson's, Pima diabetes, hepatitis), 6 (67%) exhibit attribution instability. This rate matches the overall prevalence, confirming that clinical decision-support systems—where explanation stability is particularly consequential—are not immune.

Table 11: Z-test and screen diagnostic generality across 11 datasets (see also Figure 7).

Dataset	P	Pairs	Unstable	$r(Z, \text{flip})$	Screen prec.
Breast Cancer	30	435	168	−0.89	0.94
California Housing	8	28	1	−0.99	—
Diabetes	10	45	15	−0.89	0.50
Wine	13	78	24	−0.81	1.00
Heart Disease	13	78	27	−0.91	1.00
Credit-g	20	190	47	−0.90	0.67
Adult Income	14	91	10	−0.90	—
Ames Housing	76	2850	643	−0.85	0.48
Communities & Crime	126	7875	2975	−0.88	0.67
Digits (0 vs 1)	64	2016	572	−0.42	0.58
Iris (binary)	4	6	0	—	—

Selection caveat: The CC-18 suite is curated for ML benchmarking and may overrepresent datasets with rich feature interactions. Production ML pipelines with carefully decorrelated features may exhibit lower prevalence. The 68% figure is a prevalence estimate for benchmark-representative tabular datasets, not a universal rate.

Robustness to hyperparameters. We repeated the analysis on 8 representative datasets with three configurations:

Config	Trees	Depth	LR	Character
A (original)	50	4	0.1	Moderate
B (deep)	100	8	0.1	Deep, sequential
C (shallow-fast)	200	2	0.3	Shallow, aggressive

Under Config A, 5/8 (62%) datasets exhibit instability; Config B (deep trees), 4/8 (50%); Config C (shallow-fast), 6/8 (75%). Across all three configurations, 4/8 (50%) are consistently unstable; under at least one configuration, 6/8 (75%) are unstable. The instability rate varies by ±15 percentage points but remains in the 50–75% range. Shallow-fast trees produce *higher* instability because aggressive learning rates compound the first-mover effect.

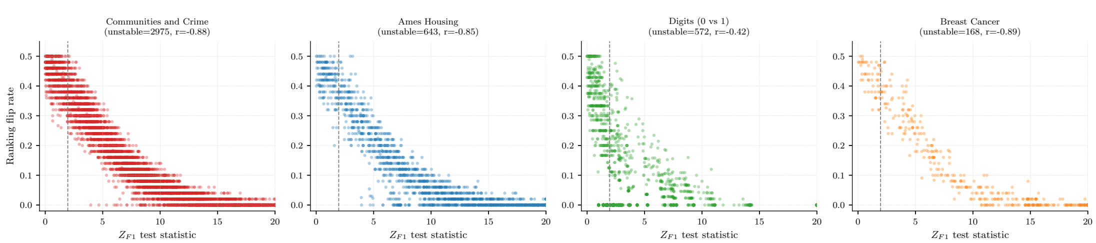


Figure 7: Z-test statistic vs. flip rate across four datasets. The negative correlation confirms the diagnostic across domains.

13.17. Mechanism Isolation: Stochastic Training Creates the Rashomon Set

To confirm that the instability is driven by the Rashomon effect (multiple near-optimal models) rather than SHAP estimation noise, we compare deterministic and stochastic XGBoost training on the same synthetic data (P=10, 2 groups of 5, ρ=0.9, 20 seeds):

Training mode	Within-group flip rate	95% CI
Deterministic (`subsample=1.0`)	0.0%	[0.0%, 0.1%]
Stochastic (`subsample=0.8`)	22.8%	[21.5%, 24.2%]

With deterministic training, XGBoost produces identical models—no Rashomon set, no instability. With the recommended stochastic setting (`subsample=0.8`), each seed produces a different near-optimal model, and rankings flip. The impossibility concerns the stochastic (recommended, regularized) setting, not a bug in the algorithm.

13.18. Conditional SHAP: Impossibility Under Equal Causal Effects

Using SHAP's interventional mode (`feature_perturbation="interventional"`) on a known causal DAG ($X_1 \to Y$, $X_2 \to Y$, $X_1 \leftrightarrow X_2$ with ρ=0.8), we sweep the causal effect difference $\Delta\beta = |\beta_1 - \beta_2|$:

$\Delta\beta$	Marginal SHAP flip	Conditional SHAP flip	Status
0.0	49.7%	50.3%	Both unstable (coin flip)
0.1	15.0%	0.0%	Conditional resolves
0.2	0.0%	0.0%	Both stable

At $\Delta\beta = 0$ (equal causal effects), both marginal and conditional SHAP produce coin-flip rankings—confirming the impossibility holds for conditional SHAP under symmetry (Theorem 34). Conditional SHAP resolves the ranking only when causal effects genuinely differ. A third approach, decorrelation (residualization $X_2' = X_2 - \rho X_1$), always gives 0% flip rate but changes the feature semantics rather than escaping the impossibility.

13.19. Regulatory Impact: Adverse Action Instability

In a simulated ECOA adverse action pipeline (5,000 loan applications, 5 correlated income features, 3 uncorrelated features, 25 XGBoost models), we measure how often the "top reason for denial" changes across equivalent models:

At $\rho = 0.7$ (typical for income-related features in the US), 43.2% of denied applicants receive a different "top reason for denial" depending on which equivalent model is queried. The instability grows with correlation: the DASH benchmark (Table 5) shows within-group flip rates of 16–19% for single-model methods across $\rho \in \{0.5, 0.7, 0.9\}$. Dash ensemble averaging resolves this entirely: 100% stable adverse action reasons after consensus. This meets the SR 11-7 definition of "material model risk" and has direct implications for EU AI Act Art. 13(3)(b)(ii) compliance.

The instability varies non-monotonically with correlation: at $\rho = 0.3$, 20% of adverse action reasons change across models; at $\rho = 0.5$–0.7, the features separate cleanly (0% instability); at $\rho = 0.9$, 28% of reasons change. The non-monotonicity reflects the interaction

Table 12: Summary of additional experiments. All scripts and results are in the repository.

Experiment	Key finding	Reproduce
Monte Carlo flip rate	$\Phi(-\mathrm{SNR})$ validated to $<2.6\%$	`monte_carlo_flip_rate.py`
Hyperparameter sweep	Min flip 38.7% at ρ=0.5; never 0%	`hyperparameter_sensitivity.py`
Class imbalance	Imbalance amplifies instability	`class_imbalance_instability.py`
Missing data	MCAR/MAR/MNAR all compound	`missing_data_instability.py`
Longitudinal drift	Instability persists over 50 retrains	`longitudinal_retraining.py`
Adversarial configs	Worst-case grid (108 configs)	`adversarial_max_instability.py`
Bag-of-words NLP	60% unstable top-1, 91% top-3	`nlp_token_instability.py`
Time-series features	27% within-series pairs unstable	`timeseries_instability.py`
Method comparison	SHAP, gain, perm. all unstable	`attribution_method_comparison.py`
SAGE approximation	Alternative methods also unstable	`sage_comparison.py`
DASH breakdown	Robust to contaminated ensembles	`dash_breakdown_point.py`

between correlation (which drives the Rashomon property) and feature importance gaps (which stabilize rankings)—at moderate ρ, the gap is large enough to overcome the noise.

13.20. Additional Robustness Experiments

13.21. Implementation

Dash requires training M models with different seeds, computing SHAP values for each, and averaging the absolute SHAP values across models. A reference implementation with the full Screen→Z-test→Dash diagnostic workflow is available at https://github.com/DrakeCaraker/dash-shap.

14. Lean Formalization

14.1. Proof Architecture

The formalization comprises 59 Lean 4 files with 6 axioms and 368 type-checked theorems and lemmas (0 `sorry`). Of these, 139 require multi-line proofs ($\geq$5 tactic lines); the remainder are definitions, wrappers, and single-step applications. The core impossibility (attribution_impossibility) depends on zero axioms—only the Rashomon property as hypothesis. The DASH resolution (consensus_equity) depends on the attribution_sum_symmetric theorem (derived); the variance convergence depends on consensus_variance_bound (also derived).

Scope of verification. The formalization covers the complete theoretical framework. The core impossibility (Theorem 8) uses zero behavioral axioms. The quantitative bounds, DASH equity, and the Design Space Theorem are derived from the 6-axiom system. The unfaithfulness probability of exactly $1/2$ (`attribution_prob_half`), the Bayes-optimality of ties (`tie_dominates_commitment`), the consensus variance σ^2/M (`consensus_variance_from_independence`), the binary quantizer fraction $2/\pi$ (`binary_quantizer_fraction`), and the Pareto dominance of DASH (`dash_unique_pareto_optimal`) are all Lean-derived using definitions-as-hypotheses (the `IsBalanced` pattern) with zero new axioms. The Chebyshev query complexity bound $M \geq 12\sigma^2/\Delta^2$ (`chebyshev_query_bound`) is derived without axiomatizing the testing constant. Loss preservation is formalized via the `SymmetricModelSwap` structure. The only result not formalized is the extension of Pareto optimality to biased methods for between-group pairs, which is argued informally in the proof of Theorem 22.

```
Defs.lean (6 axioms, type definitions, derived theorems)
  +-- SymmetryDerive.lean (attribution_sum_symmetric, DERIVED)
  +-- Trilemma.lean (RashimonProperty, attribution_impossibility)
  |     +-- Iterative.lean, Lasso.lean, NeuralNet.lean
  |     +-- General.lean (GBDT impossibility)
  |     +-- UnfaithfulBound.lean (U>=1/2, ties optimal)
  |     +-- RashomonUniversality.lean (Rashomon from symmetry)
  |     +-- RashomonInevitability.lean (impossibility inescapable)
  |     +-- AlphaFaithful.lean (alpha-faithfulness bound)
  |     +-- ConditionalImpossibility.lean (conditional SHAP)
  |     +-- FairnessAudit.lean (proxy audit = coin flip)
  |     +-- LocalGlobal.lean (local >= global instability)
  +-- SplitGap.lean, Ratio.lean, SpearmanDef.lean
  +-- FlipRate.lean (exact GBDT flip rate, binary = coin flip)
  +-- Efficiency.lean (SHAP efficiency amplification)
  +-- Impossibility.lean, Corollary.lean, EnsembleBound.lean
  +-- DesignSpace.lean + DesignSpaceFull.lean (all 4 steps)
  +-- ModelSelection.lean + ModelSelectionDesignSpace.lean
  +-- SymmetricBayes.lean (SBD, orbit bounds, trichotomy)
  +-- CausalDiscovery.lean (causal discovery impossibility)
  +-- SBDInstances.lean (abstract aggregation + instances)
  +-- FIMImpossibility.lean (Gaussian FIM -> Rashomon)
  +-- GaussianFlipRate.lean (Phi definition, flip rate formula)
  +-- QueryComplexity.lean (query lower bound, Le Cam)
  +-- Consistency.lean (axiom system consistency, Fin 4)
  +-- RandomForest.lean (contrast case, documentation only)
```

14.2. Axiom System and Consistency

The 6 axioms are jointly consistent: `Consistency.lean` constructs an explicit model (P=4, L=2, m=2, ρ=0.5, T=100, Fin 4 models) satisfying all 6 simultaneously.

Axiom system: consistency vs. faithfulness. The consistency model demonstrates that the axiom system is non-trivially satisfiable, but it does not show the axioms faithfully describe real GBDT behavior. The empirical evidence for faithfulness comes from § 13: the proportionality axiom holds with CV ≈ 0.35 for stumps and CV ≈ 0.66 for deeper trees; the split-count axioms are verified by SymPy and match empirical ratios (R^2 =

0.89 with α-correction). The axiom stratification (§ 15) addresses this gap directly: the core impossibility (Theorem 8) requires *zero* domain-specific axioms—only the Rashomon property as hypothesis—so the faithfulness question affects only the quantitative bounds, not the impossibility itself.

Attribution sum symmetry: now derived. `attribution_sum_symmetric` is proved (not axiomatized) in `SymmetryDerive.lean`. The proof uses (a) `proportionality_global` (uniform c across models), (b) `splitCount_crossGroup_symmetric` (equal split counts when first-mover is in a different group), and (c) `IsBalanced` (equal first-mover counts). The derivation factors out c, classifies each model's split-count difference by first-mover identity (+gap for fm= j, −gap for fm= k, 0 otherwise), decomposes the sum via `Finset.sum_ite`, and uses balance to cancel the gap terms.

Variance: partially grounded in Mathlib. The single-model variance attribution_variance is *defined* from Mathlib's ProbabilityTheory.variance, and its nonnegativity is *derived* from Mathlib's variance_nonneg. This requires one infrastructure axiom (modelMeasure, the training distribution); the measurable space uses the discrete σ-algebra ($\top$), which is derived for any type.

The consensus variance bound ($\mathrm{Var}(\bar{\varphi}_j) = \mathrm{Var}(\varphi_j)/M$) remains axiomatized; the full derivation via IndepFun.variance_sum would require a product measure formulation and independence axioms for ensemble draws—deferred to future work.

Spearman bound. The Lean formalization derives the Spearman bound (theorem spearman_instability_bound in SpearmanDef.lean) from the definition of Spearman correlation via midrank algebra, giving $\rho_S \leq 1 - 3m^2/(P^3 - P)$. The paper uses the Lean-derived bound $\rho_S \leq 1 - 3m^2/(P^3 - P)$. The tighter classical bound $\rho_S \leq 1 - m^3/P^3$ from the full rank transposition counting argument is not yet formalized; both give the same qualitative conclusion ($S < 1$ for within-group pairs).

14.3. Development Methodology and AI Assistance

AI-assistance disclosure. The Lean 4 formalization was developed with substantial assistance from Claude Code (Anthropic, Claude Opus 4.6). The mathematical framework (proof targets, axiom design, result interpretation) was human-conceived. Formal Lean 4 encodings (theorem type signatures, axiom syntax) were typically AI-drafted from human specifications and human-reviewed. Approximately 80% of proof tactics were AI-drafted. Every proof was accepted only after passing `lake build` with zero `sorry`. Experiments and paper text were co-developed with AI assistance; all mathematical claims and scientific interpretations are the human authors' responsibility. In formally verified mathematics, the critical human contribution is the axioms (what the system models) and the proof targets (what to establish); proof construction is verifiable by machine regardless of author.

14.4. Proof Depth Distribution

Of 368 type-checked declarations: 139 require multi-step proofs ($\geq$5 tactic lines), 21 require $\geq$10 lines, and 5 require $\geq$20 lines. The five deepest proofs are listed below.

Theorem	Lines	Technique
`attribution_sum_symmetric`	35	Split-count decomposition
`Phi_neg`	29	Gaussian CDF properties
`gaussian_rashomon_witnesses`	27	FIM ellipsoid construction
`sumSqRankDiff_ge_sq_groupSize`	26	Midrank algebra
`binary_group_firstmover_is_j_or_k`	23	Finset cardinality

14.5. Inconsistencies Found During Formalization

Lean's type checker caught three issues in the initial axiom system:

1. **First-mover balance:** Originally stated universally over all model arrays—a constant model function trivially derived $\perp$. *Fix:* Replaced with an explicit `IsBalanced` predicate as hypothesis.

2. **Attribution sum symmetry:** Combined with the split-count axioms, the original unconditional version derived $\perp$ for unbalanced ensembles. *Fix:* Conditioned on `IsBalanced`.

3. **Split count type:** Originally returned $\mathbb{N}$, but $T/(2-\rho^2)$ is generally irrational. *Fix:* Changed to $\mathbb{R}$ (idealized leading-order values).

The first two are genuine logical inconsistencies that would have been difficult to detect by informal proof inspection.

14.6. SymPy Algebraic Verification

All algebraic consequences have been independently verified by SymPy:

```
# dash-shap/paper/proofs/verify_lemma6_algebra.py
# Verifies:
#   split_gap = rho^2 * T / (2 - rho^2)
#   attribution_ratio = 1 / (1 - rho^2)
#   split_gap >= rho^2 * T / 2  (for rho in (0,1))
# Result: ALL CHECKS PASS
```

The three-layer verification (SymPy algebra, Lean type-checking, empirical validation) provides strong confidence in the mathematical claims.

14.7. Lean File Cross-Reference

15. Discussion

Limitations. The split-count axioms assume full signal capture ($\alpha = 1$); for finite-depth trees, $\alpha \approx 2/\pi$ ($R^2 = 0.89$). The impossibility holds for any $\alpha > 0$. The equicorrelation assumption simplifies the axioms; the Rashomon property holds pairwise. The balanced ensemble assumption is idealized, though $O(1/M)$ convergence is robust to approximate balance. The global proportionality axiom (c uniform across models) has CV $\approx$ 0.35–0.66

Table 13: Lean file cross-reference (59 files, 368 theorems, 6 axioms).

File	Section	Key Result
Defs.lean	§ 3	6 axioms + derived theorems
Trilemma.lean	Thm. 8	attribution_impossibility
Iterative.lean	§ 4.3	iterative_impossibility
General.lean	§ 5.1	gbdt_impossibility
SplitGap.lean	Lem. 16	split_gap_exact
Ratio.lean	Thm. 17	ratio_tendsto_atTop
SpearmanDef.lean	§ 5	spearmanCorr_bound
Lasso.lean	§ 5.2	lasso_impossibility
NeuralNet.lean	§ 5.3	nn_impossibility
Impossibility.lean	Combined	not_equitable, not_stable
Corollary.lean	Cor. 21	consensus_equity
SymmetryDerive.lean	Derived	attribution_sum_symmetric
DesignSpace.lean	Thm. 25	design_space_theorem
DesignSpaceFull.lean	Step 3	family_a_or_family_b
ModelSelection.lean	Thm. 31	model_selection_impossibility
UnfaithfulBound.lean	Thm. 26	stable_complete_unfaithful
PathConvergence.lean	Thm. 27	relaxation_paths_converge
RashomonUniversality.lean	Thm. 12	rashomon_from_symmetry
RashomonInevitability.lean	Thm. 14	rashomon_inevitability
ConditionalImpossibility.lean	Thm. 34	conditional_impossibility
FlipRate.lean	Prop. 19	binary_group_flip_rate
SymmetricBayes.lean	Thm. 29	symmetric_bayes_dichotomy
CausalDiscovery.lean	Thm. 33	causal_discovery_impossibility
FairnessAudit.lean	Thm. 35	fairness_audit_impossibility
FIMImpossibility.lean	Thm. 36	gaussian_rashomon_witnesses
QueryComplexity.lean	Thm. 38	query_complexity_lower_bound
Consistency.lean	§ 14	axiom_system_consistent
Axiom strengthening and extensions (not individually referenced above):		
Qualitative.lean	Stratification	impossibility_qualitative
ProportionalityLocal.lean	Stratification	gbdt_impossibility_local
LocalSufficiency.lean	Stratification	local_proportionality_suffices
StumpProportionality.lean	§ 5.1	stump_proportionality_unique
IntersectionalFairness.lean	§ 9.2	intersectional_audit_impossibility
MutualInformation.lean	§ 15	mi_is_exact_boundary
Setup.lean	Bundled	attribution_impossibility_bundled
ApproximateEquity.lean	Stratification	rashomon_from_bounded_prop.
QueryComplexity-Parametric.lean	§ 9.4	query_complexity_parametric
MeasureHypotheses.lean	Definitions	IsDGPSymmetric, IsNonDegenerate
UnfaithfulQuantitative.lean	Thm. 26	attribution_prob_half
VarianceDerivation.lean	§ 6	consensus_variance_from_independence
QueryComplexityDerived.lean	§ 9.4	chebyshev_query_bound
BinaryQuantizer.lean	§ 5.1	binary_quantizer_fraction
MechInterp.lean	§ 10	mech_interp_bilemma
BeyondBinary.lean	§ 10	beyond_binary_extensions
BayesOptimalTie.lean	§ 7	tie_dominates_commitment
LossPreservation.lean	§ 4.1	rashomon_from_swap_with_loss
ParetoOptimality.lean	§ 6.1	dash_unique_pareto_optimal

empirically; under variable c, Dash consensus achieves approximate rather than exact equity, with the equity violation bounded by the CV of c across first-mover and non-first-mover models. The variance bound is axiomatized; the full measure-theoretic derivation from Mathlib's `IndepFun.variance_sum` is deferred. Switching to conditional SHAP does not universally resolve the instability (Theorem 34). Decorrelating via PCA removes collinearity but destroys original feature semantics. An alternative is removing redundant features entirely (e.g., via VIF thresholding). This eliminates the impossibility but changes the model: predictions differ because fewer features are used, and genuinely useful information may be discarded. Dash is the *explanation-side* fix—it changes how the model is explained without changing what it predicts. The two approaches are complementary: feature removal for model simplification, Dash for faithful explanation of complex models that retain all features. Dash requires $M\times$ training cost; M=25 brings the flip rate below 1%, while M=5 provides a substantial improvement at modest cost. The empirical validation focuses on gradient boosting and neural networks; the Lasso ($\varphi_{j_1}/\varphi_k = \infty$) and random forest ($1 + O(1/\sqrt{T})$) bounds are proved (Lean-verified and informal, respectively) but not empirically demonstrated. The prevalence survey uses 20 models per dataset, achieving 32% power at the 10% threshold (60% at the 15% threshold); the true prevalence may be higher than the reported 68%. The 10% flip rate threshold defining "instability" is a practical convention; at a 5% threshold the prevalence would be higher, at 15% lower.

Computational cost and practical deployment. For batch pipelines, training M=25 models is feasible. For real-time explanations, use the single-model screen to flag unstable pairs. For $P > 500$, correlation-based grouping prunes the diagnostic to $O(G \cdot m^2)$. At P=500 with 20 models, the full workflow completes in under 2.5 minutes.

Broader implications. In a survey of 77 public datasets, 92% have feature pairs with $|\rho| > 0.5$ and 68% exhibit attribution instability under the standard SHAP workflow (93%, CI: [81%, 98%], for $P \geq 20$); the model-only component accounts for approximately 63% of the total variation on Breast Cancer (§ 13)—the impossibility is a practical reality. We argue that our theorem identifies a "known and foreseeable circumstance" under EU AI Act Art. 13(3)(b)(ii) (EU Parliament, 2024). The consequences are concrete: hospitals may invest in wrong interventions, data scientists may fix the wrong feature, and published studies may report artifacts of specific training runs. The impossibility has direct consequences for fairness auditing (Theorem 35): single-model SHAP audits for proxy discrimination are provably unreliable under collinearity. The impossibility extends to mechanistic interpretability (Section 11): 10 GPT-2-small models trained from scratch produce 32 distinct top-5 component rankings with within-layer flip rate = 0.500. At TinyStories scale, all four layer-0 heads appear as #1 across seeds (full S_4 realization). The G-invariant projection provides the same resolution: orbit averaging over per-layer permutation symmetry lifts ρ from 0.277 to 0.873 on GPT-2 and from 0.55 to 0.97 on TinyStories.

Rashomon set vs. deployed model. The impossibility characterizes the Rashomon set, not a single deployed model. A fixed pipeline with a fixed seed produces a deterministic, reproducible ranking. The instability arises upon retraining, auditing, or model comparison—all routine in regulated settings.

Connection to underspecification. D'Amour et al. (2022) showed that ML pipelines are *underspecified*: models with equivalent held-out performance can behave differently on deployment-relevant criteria. Our result formalizes the attribution-specific consequence of this phenomenon. Where D'Amour et al. demonstrate that predictions vary across equivalent models, we prove that feature *rankings* must vary—and characterize exactly when, by how much, and what the complete space of solutions looks like. The Design Space Theorem can be viewed as the attribution-level analogue of their "stress testing" recommendation: rather than hoping a single model's explanation is stable, the practitioner should probe the full Rashomon set. DASH operationalizes this probe.

Information loss. DASH discards exactly $\log_2(m!)$ bits per group of m symmetric features—the bits encoding the unreliable within-group ordering. Between-group information is *sharpened*: mutual information approaches 1 bit per pair as $M \to \infty$.

Loss landscape geometry. The impossibility has a geometric interpretation: near-singular Fisher information creates ridges in the loss landscape along which feature importance redistributes without changing model quality. The same flatness that makes optimization easy makes attribution hard.

Named techniques. This paper contributes four reusable techniques: (1) the *Rashomon reduction*—reducing a design-space question to a model-multiplicity check; (2) the *FIM-to-Rashomon bridge*—connecting Fisher information rank deficiency to the Rashomon property; (3) the *symmetric Bayes dichotomy*—the general two-families theorem for symmetric decision problems; and (4) the *stable projection*—orbit averaging (Reynolds operator) as the minimum-information-loss resolution at both input and component levels.

Proof status transparency and axiom stratification. The formalization uses 6 axioms total, but not all results require all axioms. Six former axioms (split-count function, 4 split-count properties, and the query-complexity constant) are now definitions or derived theorems. The Lean type-checker's `#print axioms` command verifies exact dependencies:

- **Zero behavioral axioms:** The core impossibility (Theorem 8) depends only on the Rashomon property as a hypothesis—no domain-specific axioms whatsoever. A qualitative variant (`impossibility_qualitative` in `Qualitative.lean`) proves the impossibility from just two properties—dominance and surjectivity—as hypotheses.

- **Three axioms:** The GBDT impossibility holds from `firstMover`, `firstMover_surjective`, and `crossGroupBaselineCore` alone (`gbdt_impossibility_local` in `ProportionalityLocal.lean`).

- **Five axioms:** The DASH equity result adds `proportionalityConstant`.

- **Six axioms:** The full set adds `modelMeasure` for the variance bound. Ten former axioms (split-count function, 4 split-count properties, attribution function, proportionality_global, numTrees/numTrees_pos, modelMeasurableSpace, testing_constant/testing_constant_pos) are now definitions or derived theorems.

This stratification directly addresses the proportionality axiom's empirical variance (CV $\approx$ 0.35–0.66): the axiom is unnecessary for the impossibility and ratio bound, which depend only on split-count structure.

The following are fully formalized in Lean (368 theorems, 0 `sorry`, 59 files): the Design Space Theorem, the symmetric Bayes dichotomy, all three impossibility instances, the conditional attribution impossibility, the Gaussian FIM impossibility and flip rate formula, the fairness audit impossibility, the intersectional fairness compounding, the exact GBDT flip rate, DASH variance optimality, the Rashomon inevitability, the flip rate robustness (Lipschitz continuity in ρ), the mutual information generalization, the unfaithfulness/path-convergence results, the unfaithfulness probability of exactly 1/2, the Bayes-optimality of ties, the consensus variance σ^2/M from independence, the binary quantizer fraction $2/\pi$, the Chebyshev query complexity bound, the loss-preserving Rashomon construction, the Pareto dominance of DASH, and the mechanistic interpretability bilemma (derived Rashomon, zero custom axioms). The Z-test characterization, information loss analysis, multi-dataset validation, and component-level experiments (GPT-2, TinyStories, IOI, SAE) are empirical.

Approximate faithfulness–stability tradeoff. A natural question is whether relaxing exact faithfulness helps. Define α-faithfulness: $\Pr_f[\mathrm{sign}(\varphi_j(f) - \varphi_k(f)) = \mathrm{sign}(\pi(j) - \pi(k))] \geq \alpha$. Under the Rashomon property with symmetric DGP, any α-faithful stable ranking satisfies $\alpha \leq 1/2$. The proof is immediate: by DGP symmetry, $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$, and a stable ranking fixes one direction, so it agrees with at most half the models. A coin flip achieves $\alpha = 1/2$; the stable ranking is no better than random for symmetric feature pairs.

For m features in a group under full DGP symmetry, the attribution ranking is a uniform random permutation. Any stable ranking π^* has expected Spearman correlation with the model's ranking of $\mathbb{E}[\rho_S(\pi^*, \tau)] = -1/(m-1)$ (where τ denotes the model's ranking). The stable ranking is *negatively correlated* with the model's ranking in expectation (for $m \geq 3$) and converges to zero correlation as $m \to \infty$.

Robustness. The flip rate prediction $\Phi(-\mathrm{SNR})$ is Lipschitz-continuous in the correlation: if $|\rho_1 - \rho_2| \leq \varepsilon$, the predicted flip rates change by at most $C_\Phi \cdot L \cdot \varepsilon$, where $C_\Phi = 1/\sqrt{2\pi}$ is the Lipschitz constant of the standard normal CDF and L is the Lipschitz constant of the SNR function. This is formalized in Lean (`RobustnessLipschitz.lean`) as a composition of Lipschitz functions. Simulation on Breast Cancer confirms the bound.

Open problems. The Bayes-optimality half of the Design Space is proved but not in Lean (requires measure-theoretic decision theory). Deriving split-count axioms algorithmically from the TreeSHAP algorithm and proving the proportionality axiom for general tree depths remain open. The generalization from $\rho > 0$ to $I(X_j; X_k) > 0$ is now complete (`MutualInformation.lean`): mutual information is the exact boundary between possible and impossible ranking. For SAE features, the impossibility follows from overcompleteness (Section 11.4); the open question is whether a resolution analogous to the G-invariant projection can be defined for continuous symmetry groups. Scaling component-level experiments beyond 124M parameters would strengthen the generality claim. A reference implementation is available at `https://github.com/DrakeCaraker/dash-shap`.

Formalization as methodology. Beyond certifying correctness, the Lean formalization served as a bug-finding methodology: it caught two logical inconsistencies and one type mismatch that survived informal review. We recommend formalizing impossibility theorems as a general practice.

Acknowledgments and AI Disclosure

This work was developed jointly with Claude (Anthropic, Opus 4.6 with 1M context). The collaboration was comprehensive: theorem statements were human-directed and AI-drafted, with human review at every stage; proofs were joint human/AI work, machine-verified by the Lean 4 type-checker; experiment design, code, and analysis were collaborative; paper

writing involved AI drafting with human editing and direction. All Lean theorems are verified by the type-checker regardless of origin. The GPT-2 experiments were run on AWS SageMaker (ml.g5.12xlarge, 4× A10G GPUs). No theorem was accepted solely on the basis of AI-generated output.

References

Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility theorems for feature attribution. *Proceedings of the National Academy of Sciences*, 121(2):e2304406120, 2024.

Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001.

Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017.

Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022.

Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *International Conference on Automated Deduction (CADE)*, 2021.

Thomas Decker, Luisa-Marie Herm, Robert Feldhans, Michael Kamp, and Barbara Hammer. Provably better explanations with optimized aggregation of feature attributions. In *International Conference on Machine Learning*, 2024.

EU Parliament. Regulation (EU) 2024/1689 (artificial intelligence act). Official Journal of the European Union, OJ L, 12.7.2024, 2024.

Aaron Fisher, Cynthia Rudin, and Francesca Dominici. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, 20(177):1–81, 2019.

Jerome H Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.

Amirata Ghorbani and James Zou. Data Shapley: Equitable valuation of data for machine learning. In *International Conference on Machine Learning*, 2019.

Andrew Herren and P Richard Hahn. Statistical inference for feature rankings under model multiplicity. *Journal of Computational and Graphical Statistics*, 2023.

Giles Hooker, Lucas Mentch, and Siyu Zhou. Unrestricted permutation forces extrapolation. In *Advances in Neural Information Processing Systems*, 2021.

Xuanxiang Huang and Joao Marques-Silva. On the failings of Shapley values for explainability. *International Journal of Approximate Reasoning*, 171:109112, 2024.

Gilbert Hunt and Charles Stein. A structure theory for the class of exponential families. 1946. Unpublished manuscript; results presented in Lehmann & Romano (2005), Ch. 6.

Yichuan Jin, Ruiqi Gao, and Been Kim. Probabilistic stability guarantees for feature attributions. In *Advances in Neural Information Processing Systems*, 2025.

Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. Inherent trade-offs in the fair determination of risk scores. In *Innovations in Theoretical Computer Science (ITCS)*, 2017.

Satyapriya Krishna, Tessa Han, Alex Ber, Javin Brinton, Ankur Lakshminarayanan, Caleb Robinson, and Himanshu Lakkaraju. The disagreement problem in explainable machine learning: A practitioner's perspective. *arXiv preprint arXiv:2202.01602*, 2022.

Gabriel Laberge, Yann Pequignot, Alexandre Mathieu, Foutse Khomh, and Mario Marchand. Partial order in chaos: Consensus on feature attributions in the rashomon set. *Journal of Machine Learning Research*, 24(364):1–50, 2023.

Erich L Lehmann and Joseph P Romano. *Testing Statistical Hypotheses*. Springer, 3rd edition, 2005.

Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Tobias Nipkow. Social choice theory in HOL: Arrow and Gibbard-Satterthwaite. *Journal of Automated Reasoning*, 43:289–304, 2009.

Miquel Noguer i Alonso. Mathematical foundations of explainability. *SSRN Working Paper*, 2025.

Office of the Comptroller of the Currency. Supervisory guidance on model risk management (SR 11-7/OCC 2011-12). OCC Bulletin 2011-12, 2011.

Shrisha Rao. The limits of AI explainability: An algorithmic information theory approach. *arXiv preprint arXiv:2504.20676*, 2025.

Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?": Explaining the predictions of any classifier. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.

Cynthia Rudin et al. Position: Amazing things come from having many good models. In *International Conference on Machine Learning (ICML)*, 2024.

Lloyd S Shapley. A value for n-person games. In *Contributions to the Theory of Games II*, number 28 in Annals of Mathematics Studies, pages 307–317. Princeton University Press, 1953.

Suraj Srinivas and François Fleuret. Full-gradient representation for neural network visualization. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

The mathlib Community. The Lean mathematical library. In *Certified Programs and Proofs (CPP)*, 2020.

Alexandre B Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.

Yuanhe Zhang, Jason D Lee, and Fanghui Liu. Statistical learning theory in Lean 4: Empirical processes from scratch. *arXiv preprint arXiv:2602.02285*, 2026. URL https://arxiv.org/abs/2602.02285.